THE RISE OF A NUDGE: FIELD EXPERIMENT AND MACHINE LEARNING ON MINIMUM AND FULL CREDIT CARD PAYMENTS

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The minimum payment warning, a notice that informs credit cardholders of the downside of making the minimum payment, has been described as a perverse nudge because it negatively affects those who would pay more than the minimum, presumably due to the anchoring bias. This issue is tackled in a massive field experiment by introducing a novel "statement balance warning." The experiment used email payment reminders that randomly added minimum payment or statement balance warnings. Results indicate that the messages shifted actual payment distribution depending on the warning and that payments increased when the statement balance warning was added. The analysis is combined with causal random forests to examine heterogeneous treatment effects, underlying mechanisms, and the optimum policy in different scenarios, and with an online experiment to further examine conditions in which the warnings affect payment behavior. The statement balance warning makes debtors give priority to paying a higher amount, which significantly increases payments by cardholders who are more likely to make deliberate decisions every billing cycle and improves the understanding of the consequences of not paying the statement balance. Furthermore, the optimal policy to decrease interest charges or debt delinquency indicates that cardholders should receive a combination of warning messages depending on their payment history. The results provide evidence that the statement balance warning offers a new element for financial market regulation to improve the decision-making of indebted households.

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I. INTRODUCTION

There are 2.8 billion credit cards in use around the world (Shift Processing, 2021). In the United States alone, the outstanding credit card debt totals USD $1 trillion, and more than 40% of that figure is revolving debt – i.e., it accrued interest every month (El Issa, 2019), and credit card debt keeps growing (Resendiz, 2020). This widespread use of credit cards that may offer a convenient payment method to households has also brought on financial distress for those who cannot manage their debt payment, which has been associated with increased life dissatisfaction and poorer mental health (Norvilitis et al., 2006; Fitch et al., 2011; Nelson et al., 2008; Hodson et al., 2014).[1]

Reasons for the increasing credit card indebtedness of households are not just based on liquidity constraints but also on people encountering many difficulties in understanding the information related to credit card payment (Soll et al., 2013; Giné et al., 2019). Previous work has shown that consumer characteristics, such as inattention and financial literacy, beyond local regulation, are key to understanding household financial distress (Keys et al., 2020). However, many financial literacy interventions have been proven to have no effect on people's behavior (Fernandes et al., 2014).

As a partial solution, policy regulations in several countries (e.g., United States, United Kingdom, Australia, and New Zealand) have included a mandatory minimum payment warning in credit card statements.[2] Even though these regulations aimed to reduce additional payments based on unpaid amounts, avoid a delinquent state and a vicious cycle of higher payments, several studies have shown, using administrative data or laboratory-based experiments, that *highlighting the minimum payment* may actually cause lower overall and full balance payments (Hershfield and Roese, 2015; Salisbury, 2014; Stewart, 2009; Wang and Keys, 2014; Navarro-Martinez et al., 2011; Adams et al., 2018). For this reason, the minimum payment saliency policy has been described as a perverse nudge (Wang and Keys, 2014; The Economist, 2008).

The reduction of payments due to the salience of the minimum payment has been commonly explained by the anchoring bias (Tversky and Kahneman, 1974): people's tendency to use the minimum amount as an anchor and insufficiently adjust their payment (e.g., Thaler and Sunstein, 2008; Stewart, 2009; Navarro-Martinez et al., 2011). Other research has argued that this effect may be triggered by the minimum payment acting as a target value (Bartels and Sussman, 2016); people exert effort to get to a specific amount. Regardless of the explanation — anchoring bias or target value — research shows that the minimum credit card payment drives cardholders' payment behavior even when accounting for liquidity constraints and financial distress (Guttman-Kenney et al., 2018; Keys and Wang, 2019).

[1] After conducting a systematic review, Fitch et al. (2011) recognized the challenges in finding a causal path between debt and mental health. Nonetheless, they conclude that "it would seem implausible that indebtedness makes no contribution towards poorer mental health over time" (p. 163).

[2] In the United Kingdom, for example, "Please make sure your minimum payment is credited to your account by the payment due date to avoid incurring a late fee, allowing up to 7 workings days for payment to clear. If you only make the minimum payment each month, it will take you longer and cost you more to clear your balance."

This paper examines a randomized, massive preregistered field experiment introducing a novel *statement balance warning* that indicates to debtors that they will avoid paying more interest by paying in full. Credit card debtors received an email reminder to pay their credit card a few days before it was due. The message randomly varied in 4 groups, a minimum payment warning, a statement balance warning, both warnings, and no warnings. While the minimum payment warning may push high-payers (low-payers) to decrease (increase) the fraction they pay of their credit card bill, based on the previous literature on lab experiments, a statement balance warning may encourage people who are able to pay more than the minimum to pay a higher fraction. However, the statement balance warning may discourage people who could only pay the minimum (or not offer the correct information). On the other hand, presenting both warning messages on the minimum and the statement balance may drive low and high payers into a self-selection process to benefit from both warnings.

The findings indicate that any warning message decreases the likelihood of not paying at least the minimum compared to a simple payment reminder without any warning, which reduces credit card delinquency. On average, the warnings decrease the less-than-the-minimum payments by debtors by 6.9 to 8.8 percent more than debtors who received the simple payment reminder (an absolute difference of 0.7-0.9 percentage points). However, to avoid revolving interest accruing because payments are not in full, the experiment's results highlight a significant change in payment distribution depending upon the specific warning message. First, debtors who received the statement balance warning are 1.0 - 1.1 percent more likely to pay in full than debtors who received the simple payment reminder (a difference of 0.6-0.7 percentage points). Second, despite the large sample size, there is no sizable change in the likelihood of paying in full for debtors who received the minimum payment warning, but they increased the likelihood of paying the *minimum payment* by 6.3 percent compared to the simple reminder (a difference of 0.5 percentage points).

The change in payment distribution has significant consequences on the interest charged in the following billing cycle, depending on the type of warnings they received. First, debtors who received a message including both warnings or only the minimum-payment warning reduced their delinquent interest (i.e., the interest charged for not paying at least the minimum) by 3.8 percent and by 3.2 percent more than those who received the message with only the statement balance warning, and by 8.0 and 7.4 percent more, respectively, than debtors who received the simple reminder. Also, consistent with the changes in payment distribution, there is a difference in the revolving interest charged (i.e., the interest charged for paying less than the statement balance). Debtors who received both warnings or only the statement balance warning reduced the revolving interest charges by 6.0 and 3.8 percent more, respectively, than debtors who received only the minimum warning message, and 9.0 and 6.8 percent more, respectively, than those who received the simple reminder. Therefore, changes in payment distributions due to specific warning messages impact people's additional payments.

In order to explore heterogeneous treatment effects and explain underlying mechanisms, the field experiment was combined with a recently developed machine-learning technique, causal random forests (Wager and Athey, 2018). This method uses high-dimensional functions to examine heterogeneous treatment effects, an "honesty" approach of resampling observations, and a bootstrap of covariates to avoid overfitting and thus avoid some of the shortcomings of reduced-form heterogeneity analysis. There are significant heterogeneous effects. A large fraction of cardholders increase their payments because of the statement balance warnings, and a small fraction of debtors might slightly reduce their payments in full due to the minimum payment warning, although there is no clear significant perverse effect. Different factors explain a positive and large treatment effect on payments when messages include the statement balance warning: a considerable variation in how much of their bill debtors usually pay - indicative of deliberation -, and whether they partially pay their balance. Also, for the messages with both warnings, debtors increase their payments when they see a relatively small gap between the minimum and the statement balance amounts, suggesting that debtors can prioritize one warning depending on which becomes more relevant and attainable. Finally, the results from the field experiment are replicated through an online experiment reflecting digital credit card payments that incorporates the warnings. This online experiment shows that the statement balance warning significantly increases the likelihood of paying in full, and results do not seem to be explained by financial skills, financial vulnerability, or people trying to override an otherwise incorrect quick response. Instead, the warnings follow a cognitive path by helping cardholders improve their understanding of financial information.

This study makes several contributions. First, it contributes to the literature on minimum payment salience and minimum payment warnings in financial decisions using the first well-powered randomized field experiment. The use of a *randomized* experiment overcomes the potential identification problems of secondary-data analysis studies (Besley and Case, 2000; Heckman et al., 1999; Kahn-Lang and Lang, 2020). In addition, a natural field experiment is not subject to potential social desirability or Hawthorne effects that may confound results in lab experiments compared to those conducted in the field (Galizzi and Navarro-Martinez, 2019; Schwartz et al., 2020, 2021). Furthermore, previous research has found that the anchoring bias varies from lab experiments to field experiments as lab experiments magnify the influence of anchoring (Jung et al., 2016). For example, using a before-after analysis for the US 2009 CARD Act, Agarwal et al. (2015) found no sizable change in overall payment when examining the same nudge highlighting an amount somewhat higher than the minimum payment examined in previous lab experiments. As an exemption to the previous lab and observational studies in the minimum payment domain, Seira et al. (2017) conducted a field experiment in Mexico using multiple treatments. One of their treatment messages highlighted the number of months needed for customers to pay off their debt if only the minimum was paid. They found no significant difference compared to customers who received no message whatsoever (their control condition), but, as discussed by the authors, it may be difficult to find sizable effects given the sample size (about 12,900 in each treatment arm), in addition to the lack of a control condition with no minimum payment salience. Without such a control condition, it is not possible to tease apart the effect of a payment reminder (positive, as shown in previous research (Homonoff et al., 2018;

Arenas and Schwartz, 2021)) from the effect of the minimum-payment salience/warning (potentially negative). In contrast, in this paper, the field experiment's control condition includes a message with no warnings, and it has a large sample size that increases the study's statistical power and helps examine heterogeneous treatment effects useful to public policy and scientific inquiry.

Second, this research proposes a novel solution to tackle the potentially perverse effect of minimum-payment salience highlighted in the literature and to help decrease household debt. Households that pay the minimum end up paying more interest and it is possible that they may never pay their debt. Several scholars have examined and proposed the solution of increasing the minimum payment (Jiang and Dunn, 2013; Navarro-Martinez et al., 2011). However, even with a higher minimum payment, consumers could still be less likely to pay in full because of the minimum payment's influence. Moreover, constrained borrowers who benefit from the minimum payment may end up delinquent, which would decrease wellbeing. Separate from the credit card statements, Guttman-Kenney et al. (2018) offer an interesting solution of removing the minimum payment on a Website and simulating an online payment in which choices are offered (hypothetical decisions). If a participant wanted to pay less than the minimum, a prompt appeared displaying the minimum payment. However, financial institutions may be discouraged from implementing this information flow, and in many countries, there is still a relevant percentage of people who do not pay online. The authors note that the UK's Financial Conduct Authority tried to run this study with lenders, but no one wanted to do it. The statement balance warning approach is a palatable solution because lenders acknowledge that consumers deserve to know the consequences of their actions. Moreover, the field experiment for this paper was implemented by a private financial institution, and served as the basis for a regulation proposal for all statement balances in a country with high credit card market penetration, which is discussed in Section VII.

Third, the experimental setting contributes to the literature examining goals and targets. Research has shown how the different tenets of prospect theory's value function (Kahneman and Tversky, 1979) can affect people's motivation and effort when one target is set (Heath et al., 1999). For example, people should exert greater effort for targets that are perceived as attainable and little effort for harder ones. In addition, psychology research postulates that when there is a hierarchy of goals or targets, one goal is prioritized and non-prioritized goals are ignored (Unsworth et al., 2014). Both sets of research predict that credit card payment may depend on which target values are prioritized: if cardholders care about paying the statement balance to avoid revolving interest, they will pay attention to the statement balance warning and ignore the minimum payment warning. This paper's findings suggest that cardholders who have usually paid more than the minimum are more likely to increase their payment under the statement balance warning, and those who have usually failed to pay at least the minimum are more likely to increase their payment when receiving the minimum-payment warning message. Because there is no empirical research examining this type of *dual* or *multi-level* targets in real-world behavior, this paper will be the first one to study how people prioritize two-level targets using a behavioral response that may be useful to financial decisions and other domains. For example, field research offers one interesting, popularized case using social norms in energy consumption with dual-target information (e.g., Allcott and Rogers, 2014). Utility companies inform two amounts to customers: one indicates the neighbors' average energy consumption and the other the usage of the most energy-efficient households, so they do not demotivate those consuming less than the average household (Schultz et al., 2007). However, there is no current examination of how household responses may prioritize these two targets.

Fourth, this paper contributes to the literature using causal machine learning in economics (Athey and Imbens, 2019). One crucial difference to most previous studies using causal random forests is that they use existing randomized control trials to find new heterogeneous effects (e.g., Davis and Heller, 2017). However, almost all behavioral interventions are designed to positively affect the average person.[3] In contrast, based on the perverse nudge effect of the minimum salience payment found in previous research, the field experiment in this research was designed to examine heterogeneous effects.

Finally, this paper contributes to the research on consumer financial markets, their regulation, and the role of behavioral biases (Agarwal et al., 2017; Campbell, 2016; Bhamra and Uppal, 2019). In particular, this research contributes to credit card disclosure regulation, as promoted in several countries (the Credit Card Accountability Responsibility and Disclosure Act in the U.S., Truth-in-Lending-Act in Mexico, Credit Contracts and Consumer Finance Act in New Zealand, and the United Kingdom Lending Code, among others). Campbell 2016's Richard Ely Lecture emphasizes the consequences on welfare and the economy with the rise of complicated financial information: consumers fail to perceive the consequence of their financial decisions, either because of complex information, lack of attention, hyperbolic temporal decisions, heterogeneous capabilities or hidden costs. Credit card issuers targeting less-educated consumers also present a systematic challenge to financial markets and regulation (Ru and Schoar, 2016). This paper's findings indicate that income and share of debt with other lenders are not the main drivers behind an increase in payments when cardholders receive the statement balance warning, which provides a palatable regulation for a broad base of consumers. Additionally, regulating the information communicated in the credit card billing process can produce substantial benefits without forbidding people's indebtedness or increasing costs. The statement balance and minimum payment warnings transparently explain the consequences of paying less than a certain amount, which should alleviate concerns about their implementation. This is aligned with policymakers' role of encouraging lenders to disclose borrowing costs so that consumers make more informed debt repayment decisions.

The rest of the paper proceeds as follows. Section II describes the background information on credit card debt, minimum payment, and payment information. Section III explains the methods, including the data and the experimental design. Section IV presents the results using a parametric analysis, and Section V focuses on heterogeneous treatment effects using causal random forests. Section VI examines different underlying mechanisms and presents

[3]This point has been raised by Susan Athey on several occasions (e.g., talk given in the AI & Big Data in Finance Research Forum on August 25, 2021).

additional analyses and experiments. Finally, Section VII concludes.

II. BACKGROUND INFORMATION

Credit cards are ubiquitous – nearly 3 billion around the world. Global statistics indicate that Canada is the country with the largest share of the population owning a credit card (83%), while the US (66%) and the UK (65%) are within the top 10 (Statista, 2017). In Latin America, Chile is one of the leading countries in credit card usage, with 20.1 million credit cards in a population of nearly 19 million people (Retail Financiero, 2019).

One of the most important measures related to credit card debt is the credit card delinquency rate. Delinquency is caused by missing one or more minimum payments, and it has consequences on individuals' credit scores and rampant debt. In the US alone, the 30-day delinquency has varied from 6.8% in 2009 to 1.8% at the beginning of 2021, probably due to the government financial aid in the midst of the Covid-19 Pandemic, and the 90-day delinquency rate continued increasing to 10% in 2021 (Board of Governors of the Federal Reserve System, 2021). In Chile, where this paper's main study takes place, two-thirds of the credit cards are managed by retail credit card issuers. The delinquency rate for retail credit cardholders was 8.4% in 2018, rising to 9.1% during mass protests in the last quarter of 2019. It varied from 7.9% to 12.1% during the Covid-19 pandemic, also influenced by the financial aid to Chilean families (Retail Financiero, 2021).[4]

Financial regulations in many countries have established payment information that must be presented by the credit card issuer. One key piece is the required minimum payment, which is generally an amount equal to at least the sum of interest and fees accrued at the moment of billing. However, credit card issuers have flexibility in calculating the minimum payment when it is greater than a required threshold. For example, some credit card issuers use a percentage of the statement balance (e.g., between 1% and 3%) or a minimum amount (e.g., $25) if the statement balance is less than the amount corresponding to the sum of 1% of the statement balance, interest and fees.[5] Like in many countries, in Chile, credit card issuers must have a minimum payment of at least the sum of credit card interest, except "when the [required minimum] is waived for a specified period under a promotion or offer" (MinEcon, 2012). Compared to the US and the UK, whose minimum payment ranges from 1% to 3%, credit card issuers vary their minimum payment by much more in Chile (between billing cycles and among consumers). The credit card issuer involved in the study, one of the leaders in Latin America, has an average minimum payment of 29% of the statement balance (SD = 20%; $Median$ = 24%; Mean SD_{within} = 13%). This variation depends on consumers' purchases (e.g., if the purchase was made using the credit card in retail stores), making the minimum vary from $0 to the statement balance, although this is unknown to the consumer.

[4] For this reference, the delinquency rate is calculated using loans for which payment is 30 days or more past due, divided by total loans (the numbers are very similar to the delinquency rate using the customer base of the retail credit card issuer in the field experiment of this paper, which is one of the major players in the country). Bank issuers have historically lower levels of delinquency rates.

[5] For examples of several credit card issuers in the U.S, see: https://creditcards.usnews.com/articles/how-credit-card-issuers-calculate-your-minimum-payment.

As introduced in the previous section, several countries have required credit card issuers to include a minimum payment warning. However, in the case of Chile, the current regulation does not require any warning of that sort, and there is no evidence that any credit card issuer is using one. On the other hand, on their websites, the Consumer Protection Bureau and the Financial Market Commission attempt to alert people that paying less or only the minimum may keep families in debt for a long time or from even paying their debt, even if they stop using their credit cards, although it would be venturesome to say that consumers are paying attention to this information.[6]

III. DATA AND EXPERIMENTAL DESIGN

III.A. Data and Sample Selection

In collaboration with a major Latin American retail credit card issuer, customer selection was based on operational requirements and on increasing statistical power.[7] The sample inclusion criteria considered customers who had no automatic payment at the time of the study, those who paid less than the statement balance at least twice in the previous 12 months, and those who were sent at least six emails by the retailer with an email opening rate of at least 20% in the previous six months.[8] The retailer requested the exclusion of a very small percentage of long-term delinquent accounts treated by other programs. Other sample selection criteria were merely operational: the email was uniquely assigned to one customer, and customers had to have a minimum payment of at least \$2.40 and a statement balance greater than or equal to \$12.0, so customers would have something to pay and could distinguish between the minimum payment and the statement balance.[9]

The company provided anonymized customer information using the above criteria to conduct a block randomization procedure using the minimum payment and statement balance, the percentage paid of the previous bill, and the number of times the customer paid less than the statement balance in the previous twelve months. Then for each block, individuals were randomly assigned to four experimental conditions. The sample consisted of 179,706 debtors with a total of 2,853,939 observations.[10] There are almost two years of data on most customers, from March 2018 to November 2019.[11]

[6]E.g., "The minimum payment can be a lifetime debt" from https://www.sernac.cl/portal/604/w3-article-1094.html, and https://www.cmfchile.cl/mascerca/601/w3-article-27871.html.

[7]Companies' concern about consumers' indebtedness is not uncommon. Some banks and credit card issuers provide tools to their customers (e.g., "inControl") so that they can limit their expenditures and control their shopping behavior (Lieber, 2010).

[8]A third-party firm measures email opening rates, mainly retail offers and promotions. Even though this measure is not exact because it depends on particular email providers and email settings, it may be used as a proxy for consumers opening the emails sent by the company.

[9]In American dollars adjusted by the Power Purchase Parity (PPP), using the exchange rate at the time of the experiment.

[10]Forty-seven people were excluded. Of those, 44 had a credit line exceeding \$48,114 (99.98th percentile), which may represent a business credit line, and 3 had a negative payment amount (they may have had some sort of credit). The Online Appendix shows the main results including these accounts.

[11]The company provided data for January-February 2018, but there were substantial missing values at the beginning of the year.

III.B. Procedure

Customers received two email payment reminders five and two days before their credit card payment was due. Emails were sent for one billing cycle during August and September 2019. All email subjects indicated the name of the customer: "[Name], [Credit card name] has important information for you." Previous research has shown that email opening rates improve when including people's names in the subject (Sahni et al., 2018). Also, all email bodies included a reminder sentence, "We remind you that your [card name] is due on [date]." Individuals were randomly assigned to four experimental conditions through the block randomization procedure explained above (25% to each experimental condition). The first experimental condition (the "Minimum Payment Warning" condition, or "MinW") included a sentence about the minimum payment: **"If you at least pay the minimum ([$]) before the due date, you will pay additional interest charges but avoid late fees."**, similar to the minimum payment warning sentence used in several countries. The second experimental condition (the "Statement Balance Warning" condition, or "SBalW") included a sentence about the statement balance: **"If you pay the statement balance ([$]) before the due date, you will avoid additional interest charges."** The third experimental condition ("Both Warnings" condition, or "BothW") combined the previous two experimental conditions, including the statement balance and minimum payment warnings.[12] Finally, the fourth experimental condition ("Control") was a pure reminder with no statement balance or minimum payment warnings.[13] All customers received their credit card statements in a separate communication that included the statement balance and minimum payment amounts. Statistical power calculation and the sentences were first tested in a pilot study with multiple treatments that showed that people reacted to the warnings rather than to the specific amounts (see the Pilot Study and Table A1 in the Online Appendix).[14] The experiment was preregistered at the Wharton Credibility Lab at the University of Pennsylvania (https://aspredicted.org/blind.php?x=gz9gw3).

III.C. Sample Characteristics and Balance Checks

Table I shows descriptive statistics and checks the balance across experimental conditions. All conditions present very similar values for pretreatment variables, with no sizable statistical differences. Debtors (female: 51.6%, 44.4 years old) received a credit card statement in the treatment period with a statement balance and a minimum payment of $1,949 (SD

[12]Original sentences in Spanish: "Te recordamos que el pago de tu [card name] vence el [date]." Minimum warning: "Si pagas al menos el Monto Mínimo [$] antes de la fecha de vencimiento, pagarás intereses pero evitarás pagar gastos de cobranza." Statement balance warning: "Si pagas el Monto Total Facturado [$] antes de la fecha de vencimiento, evitarás pagar intereses adicionales."

[13]The company reviewed the messages before they were launched to ensure that they were understandable. It also asked that the statement balance amount be included if the minimum payment was included, even if it was at the end of the email (so, the minimum payment was also included when adding the statement balance at the end of the email). Customers received both amounts in their monthly statements.

[14]The online experiment in Section VI also provides evidence that people react to the warning sentences and not to the amounts.

= 3,050) and \$416 (SD = 529), respectively.[15] Before the experiment, using all 2019 billing cycles before the treatment period, debtors paid an average of 75.8% of their statement balance. In addition, on average, 63.7% paid in full, 7.9% paid the minimum payment,[16] and 8.2% did not pay or paid less than the minimum payment, leading to credit card delinquency, additional interest, and late fees. This percentage is also similar to the average national delinquency rate for credit cards issued by retailers at the time of the study, as described in Section II.

IV. CHANGES IN PAYMENT BEHAVIOR

This section examines the effect of the warning messages on payment behavior, including how shifts in payment distribution affected interest payments. Equation 1 estimates the effect of the warning messages using an ordinary least square analysis with robust standard errors clustered by individual (Bertrand et al., 2004).

$$y_{it} = \alpha + \sum_j \beta_j D_{ij} \times P_t + \delta P_t + X_{it} + \mu_m + \mu_y + a_i + \varepsilon_{it} \tag{1}$$

where D_{ij} is a dummy indicator for each warning message assigned to individual i compared to the control (treatment warnings j = {Minimum, Statement Balance, Both}). P_t is a dummy variable for before and after the treatment period for billing cycle t. The specification includes controls for changes in customers' characteristics over time and before the treatment (X_{it}: company's internal credit score, income, purchases, statement balance and the ratio between the minimum payment and statement balance of individual i in billing cycle t), monthly and yearly fixed effect (μ_m and μ_y), and individual fixed effects (a_i). The error term is ε_{it}. β_j indicates the average intention-to-treatment effect of each warning message as compared to the control condition. Because of randomized assignment, these difference-in-difference estimators are unbiased (Rubin, 1974). The outcome variable (y_{it}) changes for each analysis; for payment variation, the specification uses the logarithmic transformation of payments of individual i in billing cycle t.[17] For the probability of not paying at least the minimum, which drives credit card delinquency, the outcome variable is a dummy indicator equal to 1 if individual i in billing cycle t does not pay or pays less than the minimum (0, if not). Similarly, for the probability of paying in full, the outcome is a dummy variable that indicates whether the cardholder i pays in full in billing cycle t. In these latter two cases, changes in probability are examined using a linear probability model. Another measure of interest is the percentage paid, which is the proportion of the

[15] Because of confidentiality, the payment information was re-scaled using a typical statement balance of \$1,949.28 in the United States (Salisbury and Zhao, 2020). Therefore, the statement balance and minimum payment were re-scaled by this value using debtors' actual statement balances multiplied by F, where F = $\frac{1,949.28}{(\frac{\sum(statementbalance_i)}{N})}$.

[16] These debtors paid exactly the minimum amount or up to the next nearest 10,000 Chilean pesos (approx. \$24 adjusted by PPP). If this variable considers the next nearest 100,000 Chilean pesos, the percentage of debtors paying the minimum would be 15.8%.

[17] For the amounts, ln(X+1).

statement balance paid by individual i in billing cycle t ($y_{it} \in [0,1]$). Finally, using the next couple of billing cycles after the treated billing cycle, two types of interest charged are also used as outcome variables; *revolving interest*, which accrues when debtors pay at least the minimum but less than the statement balance, and *delinquent interest*, which accrues when debtors do not pay or pay less than the minimum.

IV.A. Main payment variation and its consequences

Table II shows the average treatment effects of each condition using the control as the baseline. The first column shows that all warning messages increase payment amounts from 6.2% to 7.8% compared to the control condition, with no sizable statistical differences across warnings. The second column indicates that the warnings reduce the probability of not paying at least the minimum by 6.9% - 8.8% from a baseline of 9.7% (0.7-0.9 percentage points), with the statement balance warning only condition offering the smallest reduction, but with no sizable statistical difference from the other warnings. This result translates into a lower credit card delinquency rate because of the warnings. As a reference for these magnitudes, the delinquency rate in Chile at the time of the experiment was 8.4% for retail credit cards (Retail Financiero, 2019) and increased 3.7 percentage points in the worst moment of the COVID-19 pandemic, which was an extreme situation.

For the probability of paying in full (third column), debtors who received a message with the statement balance warning increased the likelihood of paying in full by 0.6-0.7 percentage points, which is an increase of 0.9% to 1.1% from a baseline of 62.5% (i.e., most people paid in full under the control condition), which decreased the revolving interest accrued by paying less than the statement balance. In contrast, there was no statistically significant change in the likelihood of debtors who received the minimum payment warning paying in full as compared to the simple reminder (CI95% [-0.3 pp., 0.7 pp.]), i.e., not clearly supporting previous lab evidence, although this result masks some heterogeneity that will be examined in the next section. Moreover, debtors who received the minimum payment warning only were 0.4 percentage points less likely to pay in full than those who received the statement balance warning or both warnings.

The previous results have several consequences. First, the proportion paid of bills in conditions that included the statement balance warning increased by 0.7 percentage points compared to the plain remainder and by 0.4 percentage points compared to the minimum payment warning condition (fourth column). Debtors who received the only minimum payment warning did not pay a statistically sizable different proportion of their bill compared to the control condition. A more important consequence for household finance is the interest that households ended up accruing in the billing cycle following the intervention, depending on their payment behavior because of the warnings. Table II (fifth column) shows that debtors who received a message with the statement balance warning reduced their revolving interest, which accrues on unpaid amounts above the minimum, by 6.8% and 9.0% for the statement balance and both warnings conditions, respectively, compared to debtors who received the simple reminder. Moreover, receiving the statement balance warnings reduced the revolving interest more than for debtors who received the minimum payment warnings only, by 3.8% and 6.0% for the statement balance and both warnings conditions, respectively. Furthermore, consistent with column 3 of Table II, there is no statistically significant difference between debtors who received the minimum payment warning and those who received the simple reminder on revolving interest.

Finally, Table II (last column) shows that all warning messages decreased delinquent interest, i.e., interest accruing because of non-payment or payments below the minimum in the billing cycle after the intervention. Specifically, debtors who received a message including the minimum payment warning decreased revolving interest by 7.4% for the minimum payment warning only condition and 8.0% for the both warnings condition, compared to the control condition. Under these two conditions, debtors also paid less revolving interest than debtors who received the statement balance warning only, by 3.2% and 3.8%, respectively. These results indicate that the different warnings affect payment distribution, with significant consequences on how much and what type of interest debtors end up accruing. Using both warnings simultaneously decreased the interest of people who would not have paid at least the minimum and those who do not pay in full. Using one of the warnings greatly affected the specific interest charged, depending on the focus of the warning.

IV.B. Payment distribution

To examine whether the previous result was in response to changes in payment distribution, Table III shows analyses using Equation 1, with different payment values or ranges as outcome variables. The first column shows that all warning messages reduced the probability of not paying at all compared to the control condition. Furthermore, it can be seen that changes in payment distributions as a result of the warning messages came mainly from a reduction in debtors who did not pay or who did pay but paid less than the minimum (first two columns). The question is where this reduction went and whether it depends on the specific warning message.

For debtors who received the minimum payment warning only, Table III indicates that they shifted their payment mostly toward the minimum amount (each row has to add up to 0). In particular, the third column shows that debtors who received this warning increased the likelihood of paying the minimum by 6.3% compared to the control (or 0.5 percentage points, with a baseline of 7.3%), i.e., 65.8% of the overall reduction in non-payment or in paying less than the minimum went toward paying the minimum for debtors who received the minimum warning only (0.52 pp. out of 0.79 pp.). In addition, Table III shows no significant changes in payments above the minimum for this group. This shift in payment change due to the minimum warning can also be appreciated in Panel A of Figure 1 that shows the average treatment effects of Table III.

While payment changes shifted toward the minimum for the minimum payment warning condition, payment changes in the conditions that included the statement balance warnings shifted almost entirely toward full payment. For example, for the statement balance warning condition, if the reduction in the likelihood of not paying at least the minimum is aggregated with the change in the likelihood of not paying the minimum, the reduction is 0.79 percentage points. Eighty-one percent of that reduction went toward payment in full (i.e., 0.64 pp. out of the 0.79 pp.), and the rest toward a revolving amount between the minimum payment and the statement balance (Panel B in Figure 1).

Table III and Figure 1 (Panel C) show that debtors who received both warnings behave very similarly to those in the statement balance warning condition, shifting 80% of the reduction of not paying even the minimum or paying a revolving amount towards paying in full. However, the remaining 20% went toward the minimum amount. In the both warnings condition, debtors consistently increased their likelihood of paying the minimum compared to debtors who received only the statement balance warning (statistical significance at the 10% level in this case for the linear model and at 5% for the logit model in the Online Appendix Table A2). Overall, these results show that the warning messages increased payments by shifting their distribution consistent with the specific warning content.

The last two columns in Table III show whether cardholders paid the statement balance (rounded up to the nearest CLP 10,000) or paid more than that, even though there is no economic benefit of paying more than the statement balance (in the baseline, 6.8% of debtors paid more than the statement balance).[18] There is no statistically significant difference in the likelihood of paying more than the statement balance, although, directionally, there is a small reduction for debtors who received the statement balance warning only compared to the control condition. Finally, Table A2, Table A3 and Table A4 in the Online Appendix show that the main results are robust using a logit model, different specifications, or including accounts with large credit lines.

V. CAUSAL FORESTS AND HETEROGENEOUS TREATMENT EFFECTS

Machine learning methods have recently focused on causal inference, which offers an opportunity for many economic problems (Athey and Imbens, 2017, 2019). The causal forests method in particular is a machine-learning technique recently developed to examine heterogeneous causal effects with high-dimensional functions of covariates, which allows behavioral mechanisms and optimal policies to be analyzed for multi-treatment cases (Athey and Wager, 2021). This section shows the heterogeneous treatment effect for each warning message, which is especially relevant considering previous evidence on the perverse effect of the minimum payment salience. Then it characterizes the sub-samples with the largest treatment effects for each warning message and, finally, presents an optimal policy analysis.

V.A. Heterogeneous effect

This part presents the heterogeneous effect analysis using causal random forest models, based on the Generalized Random Forests algorithm (Athey et al., 2019) to estimate the conditional average treatment effect (CATE). In regression or classification trees, covariates are used to split observations into branches to find leaves that can predict treatment effects (Breiman and Stone, 1984), random forests solve individual tree inaccuracy and overfitting

[18]This behavior may be due to the convenience of paying subsequent installments in advance.

by first aggregating trees using a bootstrap procedure with a sub-sample m of covariates for each tree.[19] Second, the algorithm selects a random subset of the sample for training to determine the model parameters and another for testing to make the estimation (Athey et al., 2019; Athey and Wager, 2019). Finally, individual tree treatment effects are weighted across the forest.

The CATE, $\tau^D(x)$, conditional on debtors' characteristics (x), is defined for each warning message, $D = \{\text{Minimum, Statement Balance, Both}\}$, as:

$$\tau^D(x) = E\left[Y_i(1) - Y_i(0)|X_i = x\right] \tag{2}$$

where $Y_i(1)$ is the potential outcome if debtor i would have been assigned to the warning message D, and $Y_i(0)$ is the potential outcome if debtor i would have been assigned to the control group. The causal forest algorithm averages individual predictions of each tree b to produce out-of-bag CATE estimates (see details in Athey et al. (2019)).[20] Wager and Athey (2019) show that these CATE estimates are consistent and asymptotically normally distributed. For further examination, observations can be ranked by their out-of-bag CATE and grouped in subgroups (e.g., quintiles) to obtain valid confidence intervals (Athey and Imbens, 2016). The conditional average treatment effect estimates within each subgroup can then be obtained using Equation 3 for each warning condition:

$$y_i = \alpha + \sum_{k=1} \tau_{kD} w_{Di} \times n_{ki} + \sum_{k=1} \vartheta_k n_{ki} + u_i \tag{3}$$

where y_i is the outcome variable, w_{Di} is 1 if cardholder i was in the warning condition D and 0 if i was in the control group. n_{ki} is a dummy variable equal to 1 if cardholder i belongs to subgroup k (0 if not). τ_{kD} is the estimated conditional average treatment effect (CATE) for the k^{th} subgroup in warning condition D compared to the control condition. This process is repeated for each warning message and for each outcome variable. This approach to characterize heterogeneous effects has been used in other domains, such as the evaluation of educational, parenting, and employment programs (Athey and Imbens, 2019; Davis and Heller, 2017; Sylvia et al., 2021).

Table IV shows the results for the causal random forests sorted by the estimated CATE using Equation 3 with different dependent variables: whether cardholders paid in full ("full payment"), whether they did not pay at least the minimum ("payment less than the minimum"), and their payment as a percentage of the statement balance ("percentage of full"), for each warning compared to the control condition. In most cases, all warning treatments show significant heterogeneity. For example, for cardholders who were in the statement balance warning only condition, the top quintile indicates a conditional average treatment effect of 2.4 percentage points compared to the control group, i.e., one-fifth of

[19] By default, in the Causal Forest algorithm, $m = \sqrt{p} + 20$, where p is the number of covariates.

[20] For each random forest, the algormithm uses 30,000 trees with 10 minimum observations per node, honesty with 0.5 splittings with the GRF 2.0.2 package in R.

the sample who received this warning increased their likelihood of paying in full by 2.4 percentage points more than debtors who received a reminder with no warning message (in relative terms, this is an increase of 4.0% in the likelihood of paying in full, as indicated in Table A5). This is almost four times the average treatment effect found in the previous section. For clarity of what follows, Table IV is discussed for each dependent variable. For additional results, Figure A1 in the Online Appendix shows the complete distribution of estimated CATE for each causal random forest, and Table A6 and Figure A2 show the results for the causal random forests using combinations of warning message conditions.

Payment in full. Table IV shows that for the conditions that included the statement balance warning, at least the top two quintiles significantly increased their likelihood of paying in full (the CATE coefficients are at least more than 1.64 times their standard errors). On the other hand, there is an enormous heterogeneity for cardholders who received the minimum payment warning only; while only the top quintile increased the likelihood of paying in full, most of the quintiles show conditional average treatment effects not sizably different from the control group, and the first quintile shows a reduction in the likelihood of paying in full (a reduction of 0.8 percentage points, or 1.1%), although this is not statistically significant even at the 10% significance level. Its magnitude is similar to the findings by Wang and Keys (2014), and it may suggest a small perverse effect of the minimum warning.[21] Table IV also includes the estimates of the best linear predictors of the CATE (Chernozhukov et al. 2018a, Chernozhukov et al. 2018b) using the out-of-bag predictions for each causal random forest. The β_1 coefficient corresponds to the conditional average treatment effect. It also indicates how well the model is calibrated (a coefficient equal to 1 represents a correct causal random forest prediction). β_2 works as an omnibus test for treatment effect heterogeneity (the null hypothesis, $\beta_2 = 0$, fails to find underlying heterogeneity). The results of this analysis reaffirm that average treatment effect predictions are well-calibrated and heterogeneity is present for the warnings that include only one message, in particular, for the minimum warning condition. The heterogeneity is weaker for the both warnings condition, given the positive result for most quintiles.[22]

Payment less than the minimum. Table IV shows that for the likelihood of not paying at least the minimum, which is associated with delinquency, the top quintiles for each warning message indicate a reduction of 1.2 to 1.5 percentage points compared to the control condition (a relative reduction of 5.8% to 15.5%). The warnings that include the minimum payment warning message significantly reduce the likelihood of not paying or of paying less than the minimum for the top three quintiles. In contrast, the statement balance warning does not show a sizable difference for the third quintile, and therefore fewer debtors may have reduced their delinquent interest in the next bill. Importantly, results suggest no indication that a relevant portion of cardholders would fail to pay at least the

[21] Wang and Keys (2014) also examined different types of disclosures available for some US cardholders, including the minimum payment warning and the payment calculation required to repay any debt in 3 years. They also found that the minimum payment warning decreased delinquency and increased the likelihood of paying the minimum.

[22] The best linear predictor of the CATE uses the out-of-bag predictions, $\hat{\tau}^{-i}(X_i)$, to estimate $Y_i = \beta_1 D_i \bar{\tau} + \beta_2 D_i \left(\hat{\tau}^{-i}(X_i) - \bar{\tau}\right) + \varepsilon_i$, where D is the warning treatment (vs. control) and $\bar{\tau}$ is the sample average of out-of-bag estimates.

minimum when receiving the statement balance warning – there are not even negative magnitudes in the CATE coefficients. Finally, the best linear projection analysis shows a statistically significant heterogeneity only for the minimum payment warning condition. In the Online Appendix, Table A6, which shows the causal random forests between warning conditions, the quintile with the largest treatment effect shows that some people would have significantly reduced their likelihood of not paying at least the minimum if they received the minimum payment warning only instead of a message with the statement balance warning.

Percentage of full. On average, cardholders in the top quintile who received the message with the statement balance warning increased the fraction paid by 1.9 to 2.2 percentage points (a relative increase between 2.9% and 3.2%). Comparing all quintiles, the average magnitudes of the CATEs for cardholders who received the minimum payment warning message are smaller than the other warning messages. All warning treatment effects show heterogeneity based on the best linear projection analysis.

V.B. Policy optimization

The previous analysis showed that the warnings did not affect all cardholders in the same way. One interesting feature of the causal random forest method is that individuals can be classified based on their characteristics to predict an optimal policy (Athey and Wager, 2021). In this case, using the results from the causal random forests for each outcome variable, the maximum out-of-bag predictions are calculated to classify cardholder i for one message (M_i). A random forest (Breiman, 2001) is then computed as a classifier using a training subset and then to predict on a test subset which predicted message ($\hat{M}_i$) cardholder i would receive based on their characteristics. This result yields the "best" predicted message each cardholder should receive for each outcome variable. However, because cardholders actually belong to a randomized experimental condition, it is possible to calculate the optimum CATE *as if* cardholders were assigned to message $\hat{M}_i$.[23] For example, using a target policy to decrease any interest charges by increasing the fraction paid, the policy optimization analysis indicates that most cardholders should receive a message with the statement balance warning: 45.3% with the statement-balance warning and 44.7% with both warnings. A small percentage (6.9%) of people should receive the minimum warning only, and the rest is predicted by the model to receive only the reminder with no warnings. Analogously, let us suppose that the optimal policy aims to reduce delinquent interest (i.e., cardholders paying at least the minimum). In this case, the analysis shows that almost every cardholder should receive a message with the minimum payment warning (42.5% the minimum payment warning and 36.4% both warnings), which is consistent with the previous heterogeneous analysis. Then, 21.0% of debtors would receive a message with the statement balance warning and 0.1% with the reminder with no warnings.

As an exercise to comprehend the role of each warning, a perverse policy that optimizes cardholders paying less than the statement balance and, therefore, accruing more interest,

[23] Across models, accuracy is over 70% on the test subset. For each condition, the accuracy of classification was over 68%, except for the control group, in which the accuracy was 55% for one of the models.

would not include any warning in the message to most of the people (54.1%), and would, interestingly, use the minimum payment warning in 32.7% of messages.[24] Finally, this policy would put the statement balance warning only in 10.6% of the messages while 2.6% would receive both warnings. This latter result indicates that some cardholders would decrease their payments if they received one warning message instead of a reminder with no warnings.

VI. UNDERLYING MECHANISMS AND ADDITIONAL RESULTS

This section examines several explanations to disentangle why the warnings affect payment behavior. In particular, the following analyses use the reduced-form analysis from Section IV, the causal random forest results from the previous section, and a new randomized online experiment.

VI.A. Advance payments

One concern about the results is that debtors may have increased their payments during the experiment by reducing their ability to pay in the following billing cycles, i.e., they advanced their payments. The firm shared data on the two billing cycles after the field experiment, and Table V shows the results for these periods using Equation 1, the dependent variables being the likelihood of paying at least the minimum, of paying in full, and the amount paid. The first post-treatment period shows that cardholders who received a message including the minimum payment warning increased their likelihood of at least paying the minimum compared to the control group that received a simple remainder; those in the minimum-payment warning condition reduced their chance of not paying or of paying less than the minimum by 2.8% (-0.4 percentage points), and those in the both warnings condition by 3.6% (-0.5 percentage points). There was no sizable change in the statement balance warning only in the post-treatment period, even at the 10% significance level, and the magnitude of its estimate is tiny (0.4%, 0.1 percentage point). In contrast, the likelihood of paying in full one billing cycle after the treatment was not sizably different from zero for any warning. Consistent with the increase in the likelihood of at least paying the minimum, the amount paid increased by 5.2% and 5.6% only for the minimum payment warning and both warnings conditions, respectively (0.1%, and not sizably different from zero for the statement balance warning condition). Two billing cycles after the experiment, there were no sizable differences for any warning in any of the outcome variables.

In summary, there are no signs that cardholders paid in advance (all the estimates either indicate a positive payment or are very close to zero). On the other hand, this evidence indicates that the effect of the warnings lasted for one period in the absence of the intervention, at least when it included the minimum payment warning message, and then it dissipated, similar to other short-lived behavioral effects (e.g., Schwartz et al. (2013)). This attenuated effect may also be due to the fact that in October 2019, Chile

[24] An organization adopting such a policy would probably not send any payment reminder.

experienced nationwide protests that may have somehow disrupted any purchase pattern as several retailers closed or changed hours of operation.[25]

VI.B. Anchoring and targets

The minimum payment warning has become one of the most prominent examples of anchoring bias (e.g., Thaler and Sunstein, 2008). However, the literature on target values would also predict that people may lean towards paying the minimum amount if the minimum payment is made salient, but through a different process. This is examined by Bartels and Sussman (2016) using a lab study and observational data, finding that people see the minimum payment salience as a target value. One of the benefits of the study in this paper is that it is possible, through a randomized field experiment, to examine whether people are following the minimum payment and statement balance warnings as *two* anchors or targets in a natural setting. Specifically, it can be examined whether debtors change their behavior to pay the minimum or the statement balance, or a little more, depending on the warning, as if they were targets, or whether they adjust around those amounts as if they were anchors. In this latter case, we should see an increase in payments below the minimum and the statement balance.

Table VI shows the average treatment effect of the warnings on paying close to the minimum and the statement balance using bins of 20 percent between payments. Based on Equation 1, the dependent variables are calculated using each payment range (=1 if payment is in that range, 0 if not). For example, the first column indicates that cardholders decreased their chance of not paying anything for all warnings (vs. the control), and the second column indicates that the likelihood of paying between any positive amount and 20% of the minimum practically did not change as a result of the warnings. Notably, almost every estimate of the top part of Table VI shows a negative magnitude for any range smaller than paying the minimum. If the minimum payment warning had acted as an anchor, we should have seen positive values around the minimum. The bottom part of the table shows the effect of all warning messages on payments around the statement balance. For example, column (10) shows the likelihood of paying an amount midway between the minimum and the statement balance, indicating no sizable change in payments. In general, there are positive estimates after the amount associated with each warning message, indicating that cardholders were more likely to pay the minimum or a little more, or the statement balance or a little more depending on the specific warning. Therefore, this analysis suggests that it is more likely that debtors saw the warnings and associated amounts as target values instead of anchors.

[25]From the New York Times: "'Chile Woke Up': Dictatorship's Legacy of Inequality Triggers Mass Protests." (https://www.nytimes.com/2019/11/03/world/americas/chile-protests.html) and from InStoreView: "Crisis in Chile: Losses and opportunities for retail (in Spanish)" (https://www.instoreview.com/crisis-en-chile-perdidas-y-oportunidades-para-el-retail).

VI.C. Previous payment behavior, the minimum payment and the statement balance

One of the key components of the experimental design of the field experiment is its dual-target setting of the minimum payment and statement balance warnings. Table VII shows differences in the average treatment effects of the warnings depending on debtors' previous payment behavior and the specific amounts in the credit card statement they received during the experiment, using the percentage of full payment as the dependent variable. This outcome variable can capture any increase in payments and showed a heterogeneous effect in all warnings based on the best linear predictor analysis.

Column 1 of Table VII shows a sub-sample of debtors who failed to pay at least the minimum in one-half or more of the months prior to the intervention, i.e., they are debtors who are more likely to become delinquent. For this sub-sample, receiving the minimum payment warning or both warnings instead of the simple reminder increased the fraction paid by 3.5 and 3.8 percentage points, respectively. On the other hand, there was no sizable difference from zero for those who received the statement balance warning.

Table VII (column 2) shows a sub-sample of debtors who most of the time paid between the minimum and the statement balance prior to the experiment (i.e., they are more likely to pay a revolving amount). They increased the fraction paid during the experiment if they received the statement balance warning or both warnings by 1.5 and 0.9 percentage points, respectively. However, there was no sizable difference from zero in this sub-sample for debtors who were in the minimum payment warning condition, consistent with the finding that this warning did not significantly reduce revolving interest. Overall, despite the relatively small sample sizes in this analysis, the results suggest that debtors follow a self-selection process by responding to specific warning messages (increasing their payments).[26]

Columns 3 and 4 of Table VII examine whether previous payment variation is affected by the warnings, using the mean absolute deviation (MAD) from the mean fraction paid. For example, a debtor who pays the same fraction of the statement balance every time will have a MAD equal to 0 and is less likely to be affected by any warning as she is following some sort of status quo. In contrast, a debtor who varies her payment fraction every month prior to the experiment probably deliberated on how much to pay, and therefore, the warnings are more likely to affect her payment decision (the maximum MAD can be 0.5 for someone who varies from paying in full to not paying at all). Using a median split of this covariate, columns 3 and 4 show that debtors who have usually changed the percentage of the statement balance they pay every month increased their payments much more due to the warnings than debtors who generally paid the same fraction, mainly when messages include the statement balance warning.

Columns 5 to 10 use the minimum payment and statement balance amounts charged in the credit card statements for the experiment as the warnings are associated with the consequences of paying these amounts. Using the median split of these amounts, the results

[26]Table A7 in the Online Appendix shows the payment distribution of these two sub-samples. It suggests that debtors who are more likely not to pay at least the minimum shift their payments toward the minimum when receiving a message with the minimum payment warning, as compared to those who received the statement balance warning only.

indicate that the effects of the minimum payment warning and the statement balance warning do not significantly vary depending on the amounts, except that the minimum payment warning seems only to increase payments of debtors who had a low minimum payment. On the other hand, the effect of the both warnings condition seems to depend not only on the specific statement balance and the minimum payment amounts but especially on their absolute difference. In particular, debtors who received both warnings increased the fraction paid relatively more when the difference between the statement balance and minimum payment was relatively small. To illustrate, columns 5 and 6 indicate that debtors who received both warnings increased the fraction paid by 1.0 percentage points (1.2%) compared to the control group, when the difference between the minimum payment and the statement balance amounts was small ($Median = \$259$), and that fraction increased by 0.4 percentage points (0.6%), although not significantly sizably different from zero, when the difference between the amounts was great ($Median = \$1{,}598$).

The previous reduced-form analyses may be susceptible to the specific median splits of the sub-samples, and any form of interaction may miss other functional relationships between variables (for completeness, Table A8 in the Online Appendix shows the results using linear and quadratic interactions). The causal random forest approach can examine whether the estimated treatment effects depend on the characteristics of debtors and bills with more flexible functions (e.g., Athey and Imbens, 2019; Davis and Heller, 2017). Using the outputs from Section V, the panels of Figure 2 show the estimated conditional average treatment effects on the y-axis, and each panel uses a different covariable in the x-axis. The dashed lines are the benchmark for the estimated conditional average treatment effect for each warning compared to the control condition. The results are mostly in line with the reduced-form analysis. For example, panel A shows that the largest treatment effects of the fraction paid are for debtors who usually partially pay their credit cards under the conditions that include a statement balance warning. Panel B shows that all warnings increase the estimated treatment effect as people varied more (MAD) how much of their statement balance they usually pay, although to a lesser degree for the minimum payment warning condition; and panel C shows that debtors with the largest (smallest) estimated treatment effects due to the both warnings condition are characterized by a small (large) gap between the minimum payment and the statement balance. Finally, Tables A9, A10 and A11 in the Online Appendix show the covariates across quintiles from Section V, sorted by conditional average treatment effects. It shows, for example, for the both warnings condition (Table A11), that a debtor from the top (bottom) quintile — the one with the largest (smallest) conditional average treatment effect — has a mean absolute deviation from the fraction paid of 24.4% (8.8%) and had a gap between the minimum payment and statement balance of $1,217 ($1,953).

In summary, these analyses suggest that, first of all, debtors are more likely to be affected by the warnings if they usually vary how much they pay. Even if all debtors included in the experiment do not have automatic payment, they still may be used to paying the same amount every month. Varying payments may be the result of deliberate decisions and, therefore, how much to pay may change depending on the warning message. This effect is separate from reminding people to pay like the message without warnings does (control condition). Secondly, debtors who were in the both warnings condition seem to pay attention to a specific warning depending on whether or not they can pay more, based on how much they used to pay and the gap between the minimum payment and the statement balance, i.e., debtors choose to prioritize a specific warning when both warnings are salient. This result offers new insights into individual decision-making in dual-target settings.

VI.D. Liquidity constraints

Similar to the previous analysis, Table VIII and Figure 3 examine whether the effects of the warnings may be associated with liquidity constraints. Consistent with prior evidence (Keys and Wang, 2019), the results show no relevant heterogeneous effects using debtors' income (columns 1 and 2 in Table VIII and Panel A of Figure 3) or the degree of indebtedness with other lenders (columns 3 and 4 in Table VIII and Panel B of Figure 3).[27] Therefore, cardholders may be increasing their payments because of the statement balance warning, mainly by reducing consumption or by using savings. Tables A9, A10 and A11 in the Online Appendix also show no clear income pattern across the quintiles sorted by the conditional average treatment effect, but they do show that debtors owing little debt to the credit card issuer in the experiment may be more affected by the both warnings message, i.e., part of the effect is coming from people who may be prioritizing repayment with one institution because of the warning.[28]

VI.E. Understanding, financial knowledge, financial vulnerability, and cognitive reflection

Previous literature has shown that other factors not captured by the field experiment may be associated with payment decisions. These factors may also be affecting the impact of the statement balance and minimum payment warnings. In particular, prior research has found a positive relationship between indebtedness caused by the effect of the minimum payment and poor financial literacy (e.g., Artavanis and Karra, 2020; Hamid and Loke, 2021), and financial vulnerability (Salisbury and Zhao, 2020). Although there is no prior research on the statement balance warning, one hypothesis is that debtors with poor financial literacy or financial vulnerability may benefit from the minimum payment warning more than a statement balance warning. On the other hand, financial literacy has been described as a long-term achievement, and messages, such as warnings, should address in-the-moment decisions (Fernandes et al., 2014). It would also be important to know whether the warnings can affect people's specific understanding of the consequences of paying less than the minimum and of paying in full.

[27] The firm uses a score of how much money is lent as a percentage share of debt in the banking system (any type of debt). This score, which goes from 0 to 2.14 (for confidentiality purposes, the score was normalized with a mean of 1.0), indicates that people owe most of their debt to other lenders when the number is almost zero and only to the firm when it is 2.14.

[28] A specific analysis of this group and their payment behavior in the subsequent billing cycle does not differ significantly from the control group. This means that debtors who have more debt with other institutions are not advancing debt payment because of the warnings.

In order to examine the relationship between the warnings and cognitive factors, an online experiment was conducted in Prolific Academic (Peer et al., 2017) with debtors from the US and the UK who had used a credit card in the previous three months. Participants (N = 400; female: 48.6%, 37.9 years old) saw a monthly credit card statement using the average monthly amount in each country and the corresponding minimum payment (currency and amounts were adjusted depending on where participants lived).[29] Then they were asked "how much would you repay?" with the following options: minimum payment, statement balance, or another amount, in which case they had to type in a number. The information and payment choices closely reflected how people worldwide see their credit card information when paying online every month. Importantly, participants were randomly assigned to four experimental conditions: minimum payment warning, statement balance warning, both warnings, and no warning (control). After people input their payment decision, they were asked how well they understood what the minimum payment and statement balance mean (from 1: "Not at all" to 7: "Very well"), how much they generally pay on the credit card (Salisbury, 2014), their average monthly credit card payment, a financial vulnerability scale (Salisbury and Zhao, 2020), financial knowledge questions (Navarro-Martinez et al., 2011), and the Cognitive Reflection Test (Frederick, 2005), which has been associated with the anchoring bias and present bias (Bergman et al., 2010; Welsh et al., 2013). Finally, participants answered demographic questions. All materials are in the Online Appendix A12.

Table IX shows the average treatment effect of each warning message on payment decisions and the understanding of the minimum payment and the statement balance. First, despite the hypothetical nature of the experiment, 16.8% of cardholders in the control group chose to pay the minimum, 33.7% between the minimum and the statement balance, and 49.5% chose to pay in full. Second, the results indicate that the warnings shifted the payment distribution. The warning that included only the statement balance warning increased the likelihood of deciding to pay in full by 18 percentage points compared to the baseline no-warning condition, and the both warnings conditions increased this likelihood by 25 percentage points. Messages including the statement balance warning also increased the fraction paid of the statement balance, replicating the result from the field experiment, although by much larger magnitudes due to the hypothetical nature of the online experiment and also because more people likely paid attention to the message at the moment of the decision. On the other hand, the minimum warning condition had no sizable effect on payment decisions compared to the no-warning condition, although it is the only condition with a positive estimate for the likelihood of paying the minimum.

The last two columns of Table IX show the changes in the understanding of the minimum payment and statement balance. First, there was a high average reported understanding of the meaning of both amounts in the baseline (above six on a 7-point scale). Second, cardholders, regardless of the type of warning, directionally increased their understanding of the minimum payment compared to control, from 0.21 to 0.24 points (significance levels at 9 to 15 percent). This suggests that any warning, even the statement balance

[29]One person was excluded because they did not complete the study.

warning, could increase people's perception of their understanding of the minimum payment. Somewhat different is the reaction of participants to their reported understanding of what the statement balance means. In this case, people who saw the statement balance warning reported an increased understanding of this concept by 0.31 (both warnings) and 0.23 (statement balance warning) points compared to the no-warning condition, with significance levels at the 4 and 12 percent, respectively. In contrast, for the statement balance understanding, the estimate of the minimum payment warning condition is much smaller (with $p = 0.41$). Despite all the limitations of this outcome variable (self-reported, ceiling effect, and statistical power), people may have improved their understanding of the statement balance warning due to the warning messages. This finding is very consistent with a study conducted with governmental institutions in Chile, which asked these questions differently: credit card owners were asked how much they would have to pay to avoid additional revolving interest. Most of them (58 percent) answered correctly (i.e., they entered the statement balance) when the credit card statement contained a statement balance warning, but only 26 percent correctly answered when they saw a credit card statement with no warnings (SERNAC, 2021).

Table A13 (A and B) in the Online Appendix examines the role of financial knowledge, financial vulnerability, and the cognitive reflection test (CRT) score. Based on the main effects of these covariates, more financially literate cardholders with higher cognitive reflection test scores would pay a larger fraction of their statement balance. Nevertheless, more importantly, there is no strong indication of a moderation of the warning treatment effects using these variables, merely suggestive evidence that messages with the minimum payment warning may be more effective for people with better financial knowledge and low cognitive reflection test scores, indicating that this warning would affect people who make a rapid decision after seeing these messages.

VII. CONCLUSION

This paper documents a new statement balance warning for credit card payments in contrast to the so-called "perverse nudge" based on the minimum payment salience. There are several results that trigger different conclusions. First, the warnings shift debtors' payment distribution depending on the specific message. The statement balance warning increases payments and reduces interest by shifting payments towards the statement balance. Compared to previous lab studies, the field experiment results on the minimum payment warning show no clear perverse effect from highlighting the minimum payment through a warning. In addition, the online study discussed in Section VI, structured in a way similar to previous lab studies, does not suggest a backfiring effect either. One explanation is that most previous lab studies focused on the time it takes to pay down the debt, and a warning may be perceived as a softer manipulation. Nevertheless, this paper finds that some cardholders are influenced to pay the minimum by the minimum warning.

Second, the quintiles based on the causal random forest show that 20 to 40 percent of the sample significantly increased their payments because of the statement balance warning. The rest already paid a large fraction of the statement balance, so little noticeable gain was possible, they were more likely to pay the same percentage of the statement balance every billing cycle or might not even pay attention to the email.[30] Warnings also seemed to be affected by the difference between the statement balance and minimum payment: it is easier to pay more if the difference between these amounts is small, while when it is great, the both warnings message may offer an option of making the minimum payment that the statement balance only warning does not. This result sheds light on how people may deal with dual-target settings, in which one of them is disregarded when it is hard to achieve. In other words, people can choose one target message if they are able to pay a higher amount. Ex-ante, it could have been predicted that the statement balance warning may even discourage low-payers from making payment, but the results suggest that people are just following the warnings as much as they can without reducing their payment. Also, the warnings seem to be treated as targets or goals that cardholders try to achieve.

Third, the cardholder behavior was very similar for the statement balance warning only and the both warnings messages. This is the case except for a difference in the likelihood of making the minimum payment when receiving the both warnings message, which is mainly driven by debtors who historically were more likely to fail to pay at least the minimum – a relatively small sub-sample in this population. The results were similar for these two conditions in the quintile analysis with the causal random forest and the online experiment. Nevertheless, the policy optimization analysis indicates a gain from using different messages to change repayment behavior, although using both warnings can be a good starting point.

Fourth, debtors who increase their payments and reduce their interest as a consequence of the statement balance warning seem not to experience liquidity constraints. So, why did they not pay more before? Recent research has explained this behavior by the credit card puzzle or co-holding puzzle (Gathergood and Weber, 2014), where debtors do not use liquid assets to repay their credit cards and reduce the costs of high interest. Consistently, Vihriälä (2020) finds that households that are more likely to co-hold are more likely to be "anchored" to paying the minimum payment, despite having the means to pay in full. As the online experiment suggests, some debtors may need a salient explanation of the consequences of not paying in full to pay more, just like the minimum payment salience did for paying the minimum.

The results also have relevant policy implications. Compared to other solutions such as increasing the minimum payment, the statement balance warning helps consumers decrease their debt without imposing a cost on the more constrained borrowers (D'Astous and Shore, 2017). In addition, while some behavioral interventions have been controversial (e.g., default options) because they depend on people not paying enough attention (Sunstein, 2019), warnings depend on the opposite, i.e., people paying attention to the information. However, one limitation is that the intervention was conducted during one billing cycle, and it may be relevant to know whether cardholders would still pay attention to the warnings if they were included every period. This may motivate future research to examine whether it is possible to have additive effects over a long period.

[30] There were tiny differences in the historic email opening rates, but in the opposite direction that one would expect; the quintile with the largest treatment effect had a slightly lower historic email opening rate.

Finally, the success of any behavioral intervention depends on its degree of scalability (Al-Ubaydli et al., 2017). The statement balance warning is highly scalable. For example, as the online experiment suggests, it is possible to use the statement balance warning in online payment settings, which are becoming more common in both developed and developing countries. Furthermore, in the aftermath of the current study, the Chilean Consumer Protection Bureau and the Ministry of Economy decided to regulate a new simplified credit card statement for all credit cards. Based on this study, one of the main features of these redesigned credit card statements is the statement balance warning. The first step in changing the regulation was to conduct an online experiment testing different versions of the redesigned credit card statement among a national sample of credit cardholders. Results showed not only that the statement balance warning increased the intention to increase payments, replicating the results of this paper, but also that it helped credit cardholders learn how much to pay to reduce revolving interest (SERNAC, 2021).

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VIII. TABLES

TABLE I
DESCRIPTIVE STATISTICS AND RANDOMIZATION BALANCE

	All	Control	MinW	SBalW	BothW	Difference p-value
	(1)	(2)	(3)	(4)	(5)	(6)
Cardholder information						
Age (years)	44.4	44.3	44.4	44.4	44.4	0.8072
Male	48.4%	48.4%	48.6%	48.4%	48.3%	0.8278
Monthly income ($)	3,645.3	3,647.4	3,651.1	3,632.8	3,649.9	0.7814
Company's internal credit score (0-100)	64.2	64.2	64.3	64.2	64.3	0.9160
Credit card statment for the treatment period						
Statement balance ($)	1,949.3	1,956.2	1,940.1	1,947	1,953.8	0.8585
Minimum ($)	415.7	414.6	414.2	417.8	416.3	0.7224
Payment behavior before the treatment period						
Previous payment ($)	1208.2	1207.3	1202.2	1202.5	1220.8	0.5065
Mean previous fraction of full	75.8%	75.7%	75.9%	75.8%	75.8%	0.8919
Previously paid in full	63.7%	63.6%	63.8%	63.8%	63.7%	0.6400
Previously paid the minimum	7.9%	7.9%	7.9%	7.8%	8.0%	0.4469
Previously did not pay at least the min	8.2%	8.3%	8.3%	8.3%	8.2%	0.7485
N cardholders	179,706	44,922	44,935	44,921	44,928	

Notes: Cardholder and payment statistics for the whole sample are in column 1 and each experimental condition in columns 2-5. Column 6 shows the p-value for the difference across conditions. Monetary values were re-scaled for reasons of confidentiality using a U.S. statement balance of $1,949.28 as reference. The credit score computed by the firm was also re-scaled to a range of 0 to 100, where a higher number is a better credit score. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.

TABLE II
WARNINGS EFFECT ON PAYMENT BEHAVIOR

Dep. Var.	ln(payment)	Less than min	Full	Percent of Full	ln(revolving interest)	ln(delinquent interest)
	(1)	(2)	(3)	(4)	(5)	(6)
MinW	0.0615***	-0.0079***	0.0020	0.0030	-0.0303	-0.0739***
	(0.0206)	(0.0018)	(0.0025)	(0.0019)	(0.0200)	(0.0155)
SBalW	0.0661***	-0.0067***	0.0064***	0.0068***	-0.0681***	-0.0422***
	(0.0207)	(0.0019)	(0.0025)	(0.0019)	(0.0200)	(0.0155)
BothW	0.0778***	-0.0085***	0.0069**	0.0070***	-0.0898***	-0.0797***
	(0.0206)	(0.0019)	(0.0025)	(0.0019)	(0.0201)	(0.0155)
p-values from pairwise comparison						
BothW vs. MinW	0.4244	0.7835	0.0443	0.0362	0.0031	0.7085
BothW vs. SBalW	0.5681	0.3317	0.8281	0.9076	0.2792	0.0156
SBalW vs. MinW	0.8226	0.4846	0.0729	0.0483	0.0595	0.0401
Baseline (Control)	11.080	0.097	0.625	0.741	4.424	1.552
R^2	0.023	0.003	0.036	0.040	0.174	0.083
N observations	2,853,939	2,853,939	2,853,939	2,853939	2,853939	2,853,939
N carholders	179,706	179,706	179,706	179,706	179,706	179,706

Notes: OLS estimates of Equation 1 using different dependent variables. Columns 1 to 4 use the payments associated with the billing cycle of the treatment period (t). Columns 5 and 6 use the credit card statement information one billing cycle after the experiment (t+1). SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE III
WARNINGS EFFECT ON PAYMENT DISTRIBUTION

Dep. Var.	No Payment (1)	Between 0 and Min (2)	Min (3)	Between Min and Full (4)	Full no Extra Payment (5)	More than Full (6)
MinW	-0.0053***	-0.0026***	0.0052***	0.0008	0.0021	-0.0001
	(0.0017)	(0.0008)	(0.0016)	(0.0023)	(0.0026)	(0.0017)
SBalW	-0.0048***	-0.0019**	-0.0012	0.0015	0.0084***	-0.0020
	(0.0017)	(0.0008)	(0.0016)	(0.0023)	(0.0026)	(0.0017)
BothW	-0.0063***	-0.0021***	0.0017	-0.0002	0.0069***	0.0000
	(0.0017)	(0.0008)	(0.0016)	(0.0023)	(0.0026)	(0.0017)
p-values from pairwise comparison						
BothW vs. MinW	0.5521	0.5177	0.0338	0.6637	0.0600	0.9534
BothW vs. SBalW	0.3689	0.7502	0.0781	0.4700	0.5798	0.2311
SBalW vs. MinW	0.7593	0.3324	0.0001	0.7734	0.0147	0.2529
Baseline (Control)	0.082	0.015	0.083	0.195	0.557	0.068
R^2	0.006	0.010	0.012	0.026	0.025	0.009
N observations	2,853,939	2,853,939	2,853,939	2,853939	2,853939	2,853,939
N carholders	179,706	179,706	179,706	179,706	179,706	179,706

Notes: OLS estimates of Equation 1 using different payment values or ranges as dependent variables. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE IV
SORTED CONDITIONAL AVERAGE TREATMENT EFFECTS PER QUINTILE

	Q1	Q2	Q3	Q4	Q5	*Best Linear Predictor* β_0	β_1
Full Payment							
MinW	-0.8	-0.1	0.1	0.4	1.4	1.01	0.76***
	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(1.37)	(0.26)
SBalW	-0.3	-0.2	0.2	1.3	2.4	1.00***	0.54***
	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(0.41)	(0.26)
BothW	0.0	0.4	1.0	1.2	1.6	0.99**	0.18
	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(0.33)	(0.27)
Payment less than the minimum							
MinW	-0.3	-0.3	-0.7	-0.9	-1.5	1.00***	0.60*
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.27)	(0.34)
SBalW	-0.2	-0.4	-0.4	-0.7	-1.2	1.00***	0.16
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.32)	(0.37)
BothW	-0.2	-0.5	-0.8	-0.9	-1.4	1.01***	0.16
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(0.26)	(0.36)
Percentage of full							
MinW	-0.3	-0.2	0.0	0.8	1.0	1.00	0.65**
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)	(0.90)	(0.27)
SBalW	-0.3	0.1	0.3	1.0	1.9	1.01***	0.46*
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)	(0.29)	(0.24)
BothW	-0.2	0.2	0.2	0.9	2.2	1.00***	0.50*
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)	(0.28)	(0.29)

Notes: Conditional average treatment effects for each quintile using the control condition as baseline. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition. For the last two columns, β_0 corresponds to the conditional average treatment effect according to the best linear predictor analysis. β_1 works as an omnibus test of treatment effect heterogeneity. One-sided heteroskedasticity-robust (HC3) SEs are between parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE V
WARNINGS POST-TREATMENT EFFECT ON PAYMENT BEHAVIOR

Period	One Billing Cycle after Treatment			Two Billing Cycles after Treatment		
Dep. Var.	Less than Min	Full	ln(payment)	Less than Min	Full	ln(payment)
	(1)	(2)	(3)	(4)	(5)	(6)
MinW	-0.0038*	-0.0007	0.0557**	-0.0014	−0.00004	-0.0121
	(0.0021)	(0.0027)	(0.0269)	(0.0019)	(0.0026)	(0.0267)
TotW	0.0005	-0.0010	0.0007	-0.0015	0.0027	0.0245
	(0.0022)	(0.0019)	(0.0027)	(0.0026)	(0.0272)	(0.0267)
BothW	-0.0049**	0.0019	0.0524*	-0.0028	-0.0017	-0.0055
	(0.0021)	(0.0027)	(0.0270)	(0.0019)	(0.0026)	(0.0267)
p-values from pairwise comparison						
BothW vs. MinW	0.6101	0.4823	0.3327	0.5347	0.9016	0.8042
BothW vs. SBalW	0.0117	0.5001	0.2740	0.0963	0.0563	0.2626
SBalW vs. MinW	0.0440	0.9790	0.9002	0.2973	0.0418	0.1710
Baseline (Control)	0.138	0.110	0.559	0.565	10.238	10.315
R^2	0.012	0.016	0.149	0.232	0.396	0.540
N observations	2,853,949	2,853,949	2,853,949	2,853,949	2,853,949	2,853,949
N cardholders	179,706	179,706	179,706	179,706	179,706	179,706

Notes: OLS estimates of Equation 1 using different dependent variables for payment behavior one billing cycle after the experiment (t+1) in columns 1 to 3, and two billing cycle after the experiment (t+2) in columns 4 to 6. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE VI
WARNINGS EFFECT ON SMALL PAYMENT RANGES

Dep Var	No Payment (1)	Payment]0,20%min] (2)	Payment]20,40%] (3)	Payment]40,60%] (4)	Payment]60,80%] (5)	Payment]80,99%] (6)	Exact Min (7)	Payment]min,20%full] (8)	Payment]20,40%] (9)
MinW	-0.0053***	0.0001	0.0001	-0.0011***	-0.0009**	-0.0009*	0.0052***	0.0010	0.0011
	(0.0017)	(0.0002)	(0.0003)	(0.0004)	(0.0004)	(0.0005)	(0.0016)	(0.0016)	(0.0014)
SBalW	-0.0048***	-0.00003	0.0002	-0.0003	-0.0007*	-0.0011**	-0.0012	0.0002	-0.0009
	(0.0017)	(0.0002)	(0.0003)	(0.0004)	(0.0004)	(0.0005)	(0.0016)	(0.0016)	(0.0014)
BothW	-0.0063***	-0.0003	0.0001	-0.0003	-0.0007	-0.0010**	0.0017	0.0014	-0.0030**
	(0.0017)	(0.0002)	(0.0003)	(0.0004)	(0.0004)	(0.0004)	(0.0016)	(0.0016)	(0.0014)
p-values from pairwise comparison									
BothW vs MinW	0.5521	0.0941	0.9689	0.0424	0.5806	0.7104	0.0338	0.7892	0.0023
BothW vs SBalW	0.3689	0.2055	0.7387	0.9166	0.9174	0.9276	0.0781	0.4241	0.1118
SBalW vs MinW	0.7593	0.6613	0.7067	0.0530	0.6521	0.6459	0.0001	0.5959	0.1498
Baseline (Control)	0.082	0.001	0.001	0.004	0.004	0.005	0.083	0.075	0.046
R^2	0.006	0.000	0.001	0.005	0.002	0.002	0.012	0.015	0.003
N observations	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939
N cardholders	179,706	179,706	179,706	179,706	179,706	179,706	179,706	179,706	179,706

Dep Var	Payment]40,60%] (10)	Payment]60,80%] (11)	Payment]80,99%] (12)	Exact Full (13)	Payment]full,120%] (14)	Payment]120,140%] (15)	Payment]140,160%] (16)	Payment]160%+ full] (17)
MinW	0.0002	-0.0002	-0.0013	0.0037	-0.0002	-0.00004	0.0001	-0.0010
	(0.0012)	(0.0010)	(0.0009)	(0.0026)	(0.0006)	(0.0005)	(0.0004)	(0.0012)
SBalW	0.0005	0.0021**	-0.0004	0.0077***	0.0003	-0.0002	-0.0006	0.0007
	(0.0012)	(0.0010)	(0.0009)	(0.0026)	(0.0007)	(0.0005)	(0.0004)	(0.0012)
BothW	0.0009	0.0007	-0.0001	0.0057*	0.0011	0.0001	-0.0006	0.0000
	(0.0012)	(0.0010)	(0.0009)	(0.0026)	(0.0007)	(0.0005)	(0.0004)	(0.0012)
p-values from pairwise comparison								
BothW vs MinW	0.5616	0.4175	0.1720	0.4438	0.0487	0.8444	0.0595	0.4039
BothW vs SBalW	0.7219	0.1826	0.8031	0.4431	0.2247	0.5283	0.9864	0.5878
SBalW vs MinW	0.8225	0.0316	0.2697	0.1245	0.4469	0.6643	0.0596	0.1686
Baseline (Control)	0.033	0.024	0.017	0.534	0.009	0.005	0.003	0.030
R^2	0.002	0.001	0.001	0.022	0.000	0.000	0.000	0.010
N observations	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939
N cardholders	179,706	179,706	179,706	179,706	179,706	179,706	179,706	179,706

Notes: OLS estimates of Equation 1 using different payment ranges as dependent variables. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE VII

WARNINGS EFFECT BASED ON PREVIOUS PAYMENT BEHAVIOR, THE MINIMUM PAYMENT AND THE STATEMENT BALANCE

Subsample	*Likely Paid Less than Min*	*Between Min and Full*	*Low MAD Percentage of Full*	*High MAD Percentage of Full*	*Small Diff between Min and Stat. Bal.*	*Large Diff between Min and Stat. Bal.*	*Low Statement Balance*	*High Statement Balance*	*Low Min*	*High Min*
Dep. Var.	Percentage of Full (1)	Percentage of Full (2)	Percentage of Full (3)	Percentage of Full (4)	Percentage of Full (5)	Percentage of Full (6)	Percentage of Full (7)	Percentage of Full (8)	Percentage of Full (9)	Percentage of Full (10)
MinW	0.0354**	0.0019	0.0011	0.0048	0.0034	0.0029	0.0031	0.0032	0.0050*	0.0011
	(0.0172)	(0.0050)	(0.0018)	(0.0033)	(0.0028)	(0.0026)	(0.0028)	(0.0025)	(0.0029)	(0.0025)
SBalW	0.0184	0.0148***	0.0022	0.0113***	0.0081***	0.0053*	0.0075***	0.0061**	0.0087***	0.0050**
	(0.0174)	(0.0051)	(0.0018)	(0.0033)	(0.0028)	(0.0026)	(0.0028)	(0.0025)	(0.0029)	(0.0025)
BothW	0.0377**	0.0090*	0.0040**	0.0097***	0.0104***	0.0037	0.0085***	0.0057**	0.0101***	0.0041
	(0.0175)	(0.0051)	(0.0018)	(0.0033)	(0.0028)	(0.0026)	(0.0028)	(0.0025)	(0.0029)	(0.0025)
p-values from pairwise comparison										
BothW vs MinW	0.8917	0.1661	0.0989	0.1459	0.0113	0.7600	0.0545	0.3351	0.0752	0.2388
BothW vs SBalW	0.2669	0.2548	0.3156	0.6330	0.4062	0.5355	0.7242	0.8518	0.6154	0.7046
SBalW vs MinW	0.3200	0.0107	0.5241	0.0536	0.0890	0.3539	0.1173	0.2495	0.2043	0.1182
Baseline (Control)	0.506	0.489	0.846	0.638	0.846	0.637	0.829	0.653	0.771	0.711
R^2	0.078	0.136	0.044	0.051	0.027	0.065	0.024	0.067	0.023	0.064
N observations	65,213	429,016	1,449,756	1,404,183	1,332,393	1,521,546	1,317,171	1,536,768	1,312,390	1,541,549
N cardholders	5,026	25,793	89,435	90,271	89,851	89,855	89,838	89,868	89,766	89,940

Notes: OLS estimates of Equation 1 using the percentage of the statement balance paid ("Percentage of Full") as dependent variable. Each column is a different subsample: column 1 shows debtors who usually failed to pay at least the minimum before the experiment, column 2 those who usually paid between the minimum and the statement balance (i.e., a revolving amount) before the experiment, column 3 (column 4) the debtors with a relatively low (high) mean absolute deviation from the fraction paid, column 5 (column 6) the debtors who saw a relatively small (large) difference between the minimum payment and the statement balance, column 7 (column 8) the debtors who saw a relatively high (low) statement balance, and column 9 (column 10) the debtors who saw a relatively high (low) minimum payment amount. SEs are between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE VIII
WARNINGS EFFECT BASED ON
INCOME AND SHARE OF DEBT LEVELS

Subsample	*Low income*	*High income*	*Low score share of debt*	*High score share of debt*
Dep. Var.	Percentage of Full	Percentage of Full	Percentage of Full	Percentage of Full
	(1)	(2)	(3)	(4)
MinW	0.0013	0.0048*	0.0045	0.0016
	(0.0027)	(0.0027)	(0.0029)	(0.0025)
SBalW	0.0085***	0.0051*	0.0076***	0.0061**
	(0.0026)	(0.0027)	(0.0029)	(0.0025)
BothW	0.0054**	0.0088***	0.0090***	0.0051**
	(0.0027)	(0.0027)	(0.0029)	(0.0025)
p-values from pairwise comparison				
BothW vs MinW	0.1201	0.1463	0.1217	0.1557
BothW vs SBalW	0.2437	0.1795	0.6332	0.6767
SBalW vs MinW	0.0065	0.9238	0.2846	0.0666
Baseline (Control)	0.760	0.722	0.710	0.772
R^2	0.042	0.043	0.038	0.044
N observations	1,389,575	1,464,364	1,391,292	1,462,647
N cardholders	89,654	90,052	89,833	89,873

Notes: OLS estimates of Equation 1 using the percentage of the statement balance paid (“Percentage of Full”) as dependent variable. Each column is a different subsample: column 1 (column 2) shows debtors with a relatively low (high) income, and column 3 (column 4) debtors with a relatively low (high) share of their debt with the lender. SEs are between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

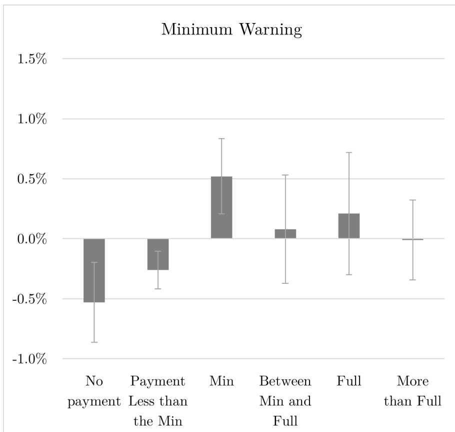
TABLE IX
WARNINGS EFFECT ON PAYMENT DECISIONS AND UNDERSTANDING
THROUGH AN ONLINE EXPERIMENT

Dep. Var.	Min (1)	Between Min and Full (2)	Full (3)	Percentage of Full (4)	Understanding Minimum (5)	Understanding Statement Balance (6)
MinW	0.028	-0.027	-0.000	0.004	0.207	0.124
	(0.051)	(0.056)	(0.068)	(0.060)	(0.146)	(0.149)
SBalW	-0.008	-0.177***	0.185***	0.154***	0.243*	0.233
	(0.051)	(0.059)	(0.067)	(0.059)	(0.145)	(0.148)
BothW	-0.080	-0.180***	0.260***	0.222***	0.233	0.311**
	(0.050)	(0.059)	(0.067)	(0.059)	(0.144)	(0.147)
p-values from pairwise comparison						
BothW vs MinW	0.035	0.011	0.000	0.000	0.862	0.211
BothW vs SBalW	0.156	0.958	0.265	0.247	0.944	0.596
SBalW vs MinW	0.484	0.013	0.007	0.012	0.809	0.470
Baseline (Control)	0.170	0.337	0.495	0.570	6.277	6.267
R^2	0.012	0.038	0.055	0.051	0.009	0.013
N observations	400	400	400	400	400	400

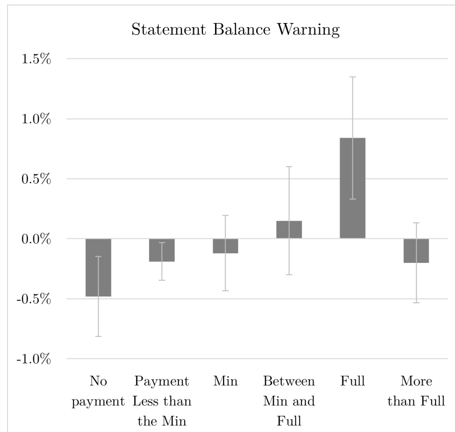
Notes: OLS estimates of Equation 1 using different dependent variables in the online experiment. Columns 1 to 4 use the payments decisions. Columns 5 and 6 use self-reported measures of understanding the minimum payment and the statement balance. SEs are between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

IX. FIGURES

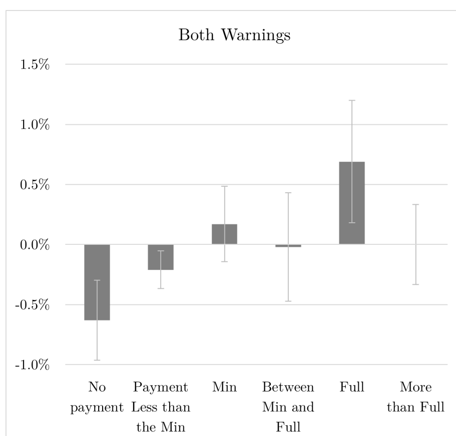
FIGURE 1. CHANGES IN PAYMENT DISTRIBUTION (WITH THE CONTROL CONDITION AS THE BASELINE)



(a) Panel A



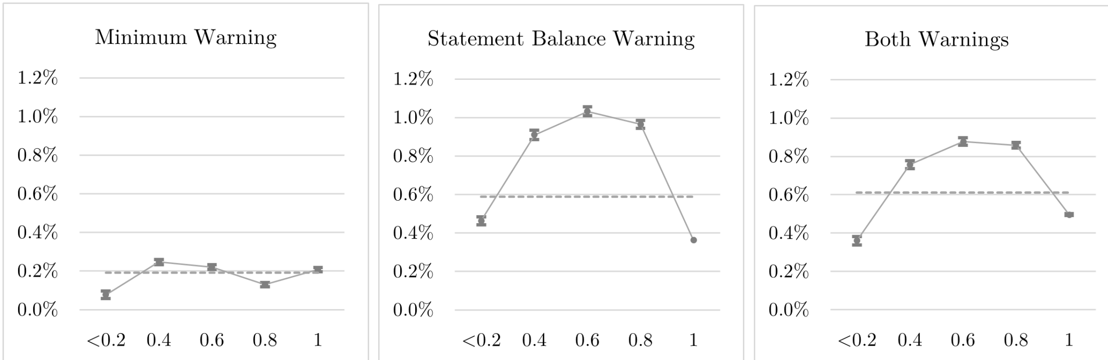
(b) Panel B



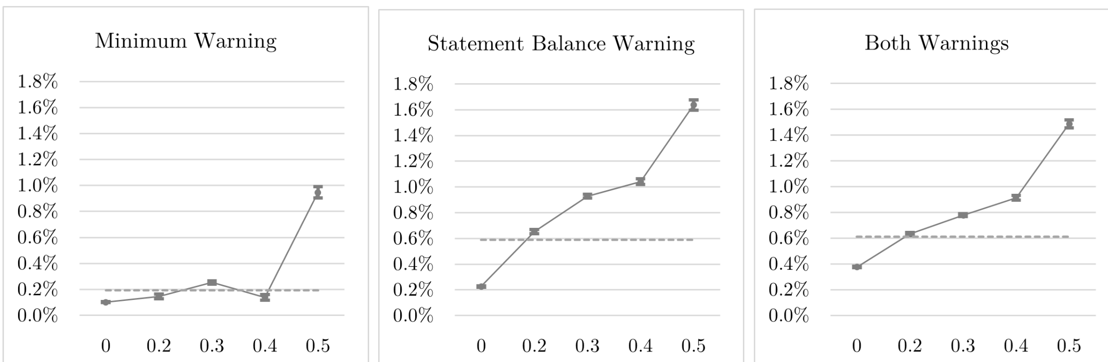
(c) Panel C

Notes: Each panel contains the average treatment effects for each warning, where the baseline is the control condition, on the likelihood of each of the possible payments (from Table III). Error bars indicate 95% confidence intervals.

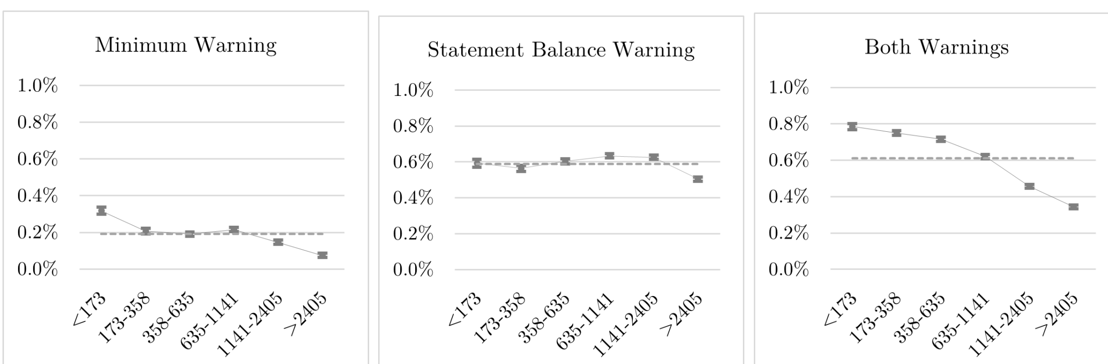
Figure 2. WARNINGS ESTIMATED CATE BASED ON PREVIOUS PAYMENT BEHAVIOR, THE MINIMUM AND THE STATEMENT BALANCE AMOUNTS



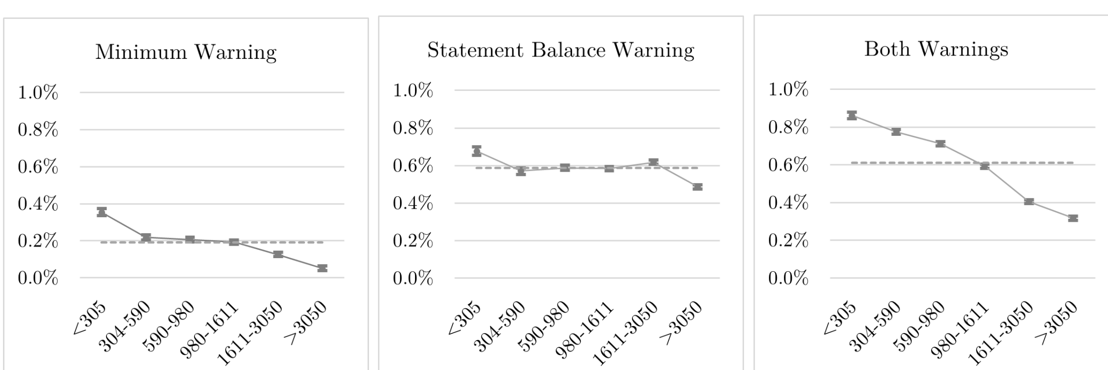
(a) Previous Percentage of Full (%)



(b) MAD Percentage of Full



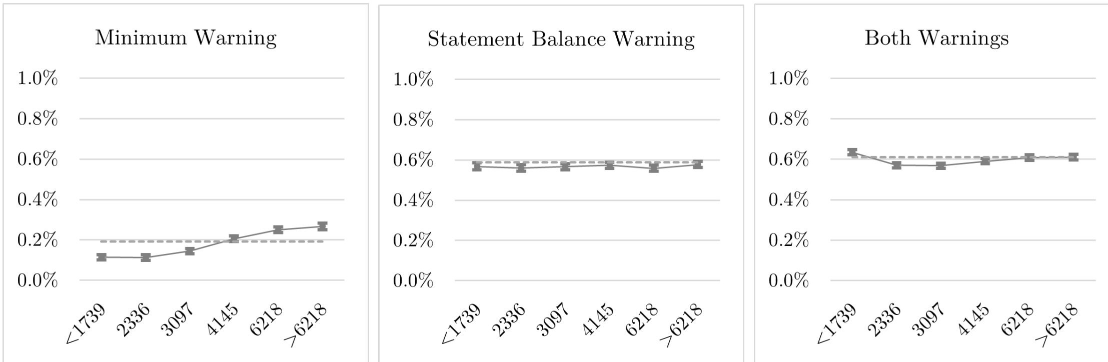
(c) Difference between the minimum and the statement balance amounts ($)



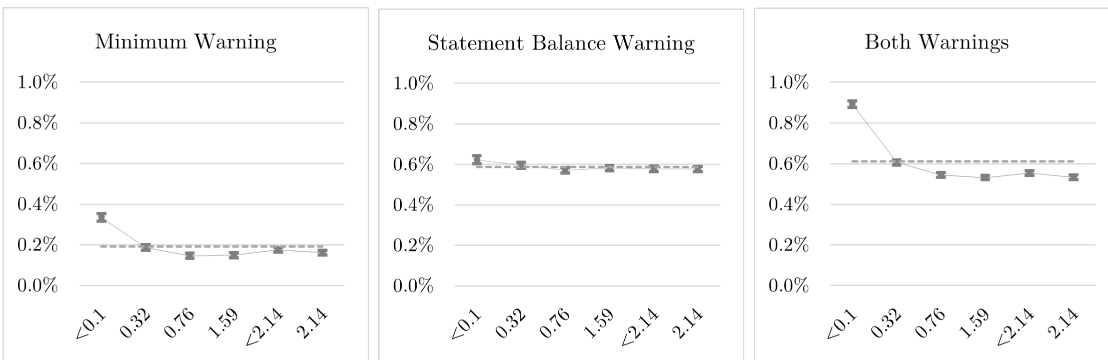
(d) Statement Balance ($)

Notes: This Figure shows the estimated conditional average treatment effects on the fraction paid on the y-axis, and each panel uses a different covariable in the x-axis. The dashed lines are the benchmark for the estimated conditional average treatment effect for each warning compared to the control condition. Error bars indicate 95% confidence intervals.

Figure 3. WARNINGS ESTIMATED CATE BASED ON INCOME AND SHARE OF DEBT



(a) Income



(b) Share of Debt Score

Notes: This Figure shows the estimated conditional average treatment effects on the fraction paid on the y-axis, and each panel uses a different covariable in the x-axis. Panel (a) uses debtors' income (in US dollars) re-scaled using a statement balance of $1,949.28. Panel (b) uses a re-scaled share of debt (0: people who have all their debt with other lenders, and 2.14: they have all their debt with the firm). The dashed lines are the benchmark for the estimated conditional average treatment effect for each warning compared to the control condition. Error bars indicate 95% confidence intervals.

APPENDIX (FOR ONLINE PUBLICATION)

A1. ***Pilot Study***

The following study was part of a different study examining the role of payment reminders with the same Latin American credit card issuer as in the field experiment in this paper but conducted more than two years before. There were seven treatment conditions sent by email and a control condition.[31] The sample size varied from 20 to 30 thousand for each treatment condition, and it was 251 thousand for the control.

- Simple reminder: "We remind you that your [card name] is due on [date]."
- Full amount only: [Simple reminder] + statement balance ($) amount in bold.
- Statement balance warning (benefit framing): [simple remainder] + statement balance ($) amount in bold + the warning: "If you pay the statement balance before the due date, you will avoid additional interest charges." [same as in the field experiment]
- Statement balance warning (cost framing): [simple reminder] + statement balance ($) amount in bold + the warning: "If you pay less than the statement balance, your outstanding balance will accrue interest charges in your next billing cycle."
- Statement balance warning (cost framing) and adding minimum: [simple reminder] + statement balance ($) amount in bold + minimum ($) amount in bold + the warning: "If you pay less than the statement balance, your outstanding balance will accrue interest charges in your next billing cycle."
- Placebo: "Remember these tips and protect your password." [and a list of standard tips]
- Control condition: People did not receive any email.

Table A1 shows the average treatment effects from linear regression models using each experimental condition as an independent variable where the baseline was the control condition. The dependent variables were the likelihood of paying in full, not paying at least the minimum, and the fraction of the statement balance.

The results indicate that all warning messages increase the likelihood of paying in full, reduce the likelihood of not paying at least the minimum, and increase the percentage of the fraction paid compared to the control group. In particular, the most successful statement balance warning message is the one framed as a benefit, and therefore that one was used for this paper, increasing the likelihood of paying in full by 2 percent (1.3 percentage points)

[31] In addition, reminders were sent at different times before the payment due date, but this factor is not analyzed here.

more than the control group. Secondly, the condition using only the statement balance amount and no warning ("Full amount only") is not statistically significantly different from the control group. Third, a simple reminder by itself increases the likelihood of paying in full by 1.1 percent (0.7 percentage points) and, given the null result from the placebo, indicates that the effect is coming from reminding people and not from the contact from the firm.

TABLE A1
AVERAGE TREATMENT EFFECTS OF THE PILOT STUDY

Dep. Var.	Full	Less than Min	Percentage of Full
	(1)	(2)	(3)
Placebo	0.0002	-0.0016	0.0001
	(0.0030)	(0.0022)	(0.0025)
Simple reminder	0.0071**	-0.0052**	0.0079***
	(0.0029)	(0.0021)	(0.0024)
SBalW (benefit framing)	0.0131***	-0.0070***	0.0107***
	(0.0028)	(0.0021)	(0.0023)
SBalW (cost framing)	0.0087***	-0.0073***	0.0082***
	(0.0028)	(0.0021)	(0.0023)
SBalW (cost framing) & adding min.	0.0030	-0.0051**	0.0050**
	(0.0028)	(0.0021)	(0.0023)
Full amount only	0.0035	-0.0034	0.0030
	(0.0035)	(0.0025)	(0.0028)
R^2	0.0001	0.0001	0.0001
N observations	411,309	411,309	411,309

Notes: SEs between parenthesis.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A2
WARNINGS EFFECT ON MAIN PAYMENT DISTRIBUTION USING A LOGIT MODEL

Dep. Var.	Less than Min	Min	Full
	(1)	(2)	(3)
MinW	-0.0860***	0.0825***	0.0150
	(0.0236)	(0.0246)	(0.0154)
SBalW	-0.0706***	-0.0141	0.0369**
	(0.0236)	(0.0251)	(0.0154)
BothW	-0.0980***	0.0441*	0.0404***
	(0.0237)	(0.0248)	(0.0154)
ll	-50,985.808	-47,627.145	-99,445.711
N observations	179,706	179,706	179,706

Notes: Logit model estimates for the treatment period using the main payment values or ranges as dependent variables. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A3

WARNINGS EFFECT ON MAIN PAYMENT DISTRIBUTION

Dep. Var.	Less than Min	Less than Min	Less than Min	Min	Min	Min	Full	Full	Full
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
MinW	-0.0079***	-0.0079***	-0.0079***	0.0050***	0.0049***	0.0052***	0.0027	0.0027	0.0020
	(0.0018)	(0.0018)	(0.0018)	(0.0017)	(0.0017)	(0.0016)	(0.0026)	(0.0026)	(0.0025)
SBalW	-0.0066***	-0.0066***	-0.0067***	-0.0012	-0.0012	-0.0012	0.0066***	0.0066**	0.0066***
	(0.0019)	(0.0019)	(0.0019)	(0.0016)	(0.0016)	(0.0016)	(0.0026)	(0.0026)	(0.0025)
BothW	-0.0083***	-0.0083***	-0.0085***	0.0015	0.0015	0.0017	0.0075***	0.0075***	0.0069***
	(0.0019)	(0.0019)	(0.0019)	(0.0016)	(0.0016)	(0.0016)	(0.0026)	(0.0026)	(0.0025)
Time effects	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
X_s	No	No	Yes	No	No	Yes	No	No	Yes
R^2	0.000	0.001	0.003	0.000	0.003	0.012	0.000	0.002	0.036
N observations	2,853,949	2,853,949	2,853,939	2,853,949	2,853,949	2,853,939	2,853,949	2,853,949	2,853,939
N cardholders	179,706	179,706	179,706	179,706	179,706	179,706	179,706	179,706	179,706

Notes: OLS estimates of Equation 1 using different main payment values or ranges as dependent variables. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A4
WARNINGS EFFECT ON MAIN PAYMENT DISTRIBUTION INCLUDING ALL ACCOUNTS

Dep. Var.	Less than Min	Min	Full
	(1)	(2)	(3)
MinW	-0.0082***	0.0052***	0.0025
	(0.0019)	(0.0016)	(0.0025)
SBalW	-0.0073***	-0.0013	0.0063**
	(0.0019)	(0.0016)	(0.0025)
BothW	-0.0090***	0.0019	0.0063**
	(0.0019)	(0.0016)	(0.0025)
R^2	0.006	0.013	0.033
N observations	3,112,110	3,112,110	3,112,110
N cardholders	179,753	179,753	179,753

Notes: OLS estimates of Equation 1 using different main payment values or ranges as dependent variables. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A5
RELATIVE CHANGE IN CONDITIONAL AVERAGE TREATMENT EFFECTS PER QUINTILE

	Q1	Q2	Q3	Q4	Q5
Full Payment					
MinW	-1.1	-0.1	0.2	0.7	2.3
	69.0	*59.0*	*70.5*	*51.7*	*62.7*
	-0.8	*-0.1*	*0.1*	*0.4*	*1.4*
SBalW	-0.5	-0.3	0.2	2.1	4.0
	68.8	*57.4*	*64.9*	*62.8*	*58.8*
	-0.3	*-0.2*	*0.2*	*1.3*	*2.4*
BothW	0.0	0.8	1.5	1.7	2.6
	65.8	*50.2*	*67.0*	*69.8*	*59.7*
	0.0	*0.4*	*1.0*	*1.2*	*1.6*
Payment less than the minimum					
MinW	-15.5	-4.0	-11.3	-7.9	-3.3
	9.9	*21.1*	*5.8*	*4.1*	*7.6*
	-1.5	*-0.9*	*-0.7*	*-0.3*	*-0.3*
SBalW	-11.8	-3.2	-8.3	-4.4	-5.9
	10.1	*21.6*	*4.7*	*8.3*	*3.6*
	-1.2	*-0.7*	*-0.4*	*-0.4*	*-0.2*
BothW	-5.8	-8.5	-18.7	-15.0	-3.8
	24.0	*10.9*	*4.4*	*3.3*	*5.6*
	-1.4	*-0.9*	*-0.8*	*-0.5*	*-0.2*
Percentage of full					
MinW	-0.4	-0.2	0.0	1.0	1.4
	63.9	*77.9*	*77.8*	*81.0*	*70.1*
	-0.3	*-0.2*	*0.0*	*0.8*	*1.0*
SBalW	-0.4	0.2	0.4	1.5	2.9
	75.5	*79.7*	*82.2*	*67.6*	*65.7*
	-0.3	*0.1*	*0.3*	*1.0*	*1.9*
BothW	-0.2	0.2	0.3	1.2	3.2
	80.4	*79.8*	*70.1*	*73.3*	*67.0*
	-0.2	*0.2*	*0.2*	*0.9*	*2.2*

Notes: The first row shows the relative change in the conditional average treatment effect (CATE) for each quintile. The second row (smaller font and italics) shows the the baseline from the control group. The third row (smaller font and italics) is the CATE in percentage points as shown in Table IV. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.

TABLE A6
SORTED CONDITIONAL AVERAGE TREATMENT EFFECTS PER QUINTILE BETWEEN WARNINGS

	Q1	Q2	Q3	Q4	Q5	*Best Linear Predictor* β_0	β_1
Full Payment							
MinW (vs. BothW)	-2.0	-1.2	-0.7	0.5	1.2	1.01**	-0.02
	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(0.42)	(0.31)
MinW (vs. SBalW)	-1.2	-0.6	-0.5	0.0	0.3	1.00*	-0.34
	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(0.57)	(0.29)
BothW (vs. SBalW)	-1.1	-0.6	0.4	0.5	0.6	1.05	-0.25
	(0.7)	(0.7)	(0.7)	(0.7)	(0.7)	(1.88)	(0.29)
Payment less than the minimum							
MinW (vs. BothW)	-0.9	-0.3	-0.2	0.9	1.1	1.21	0.50*
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(4.92)	(0.26)
MinW (vs. SBalW)	-0.8	-0.5	-0.3	0.3	0.7	1.00	0.09
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(1.93)	(0.42)
BothW (vs. SBalW)	-0.2	0.1	0.2	0.4	0.6	1.02	-0.14
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)	(1.28)	(0.42)
Percentage of full							
MinW (vs. BothW)	-1.7	-0.6	-0.2	0.1	0.4	1.01**	0.29
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)	(0.38)	(0.29)
MinW (vs. SBalW)	-0.7	-0.6	-0.3	-0.2	-0.2	1.00**	-0.23
	(0.5)	(0.5)	(0.5)	(0.5)	(0.5)	(0.43)	(0.28)
BothW (vs. SBalW)	-0.4	-0.1	0.1	0.2	0.2	1.00	0.31
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)	(4.23)	(0.28)

Notes: Conditional average treatment effects for each quintile between warning conditions. The baseline is indicated as the second group after "vs." SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition. For the last two columns, β_0 corresponds to the conditional average treatment effect from the best linear predictor analysis. β_1 works as an omnibus test for treatment effect heterogeneity. One-sided heteroskedasticity-robust (HC3) SEs are between parentheses .
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A7
WARNINGS EFFECTS ON PAYMENT DISTRIBUTION
BASED ON PREVIOUS PAYMENT BEHAVIOR

Panel A - Likely paid less than the Min				
	Less than	Min	Between	Full
Dep Var	Min		Min and Full	
	(1)	(2)	(3)	(4)
MinW	-0.0479**	0.0229*	0.0100	0.0157
	(0.0212)	(0.0132)	(0.0157)	(0.0188)
SBalW	-0.0193	-0.0007	-0.0070	0.0253
	(0.0212)	(0.0127)	(0.0153)	(0.0189)
BothW	-0.0506**	0.0155	0.0137	0.0212
	(0.0213)	(0.0130)	(0.0157)	(0.0191)
p-values for pairwise comparison				
BothW vs MinW	0.8991	0.5834	0.8131	0.7691
BothW vs SBalW	0.1352	0.2141	0.1776	0.8252
SBalW vs MinW	0.1699	0.0729	0.2681	0.6034
Baseline(Control)	0.364	0.083	0.152	0.401
R^2	0.107	0.010	0.033	0.051
N observation	65,213	65,213	65,213	65,213
N cardholders	5,026	5,026	5,026	5,026
Panel B - Likely paid between Min and Full				
	Less than	Min	Between	Full
Dep Var	Min		Min and Full	
	(1)	(2)	(3)	(4)
MinW	-0.0084*	0.0001	0.0139*	-0.0052
	(0.0048)	(0.0053)	(0.0084)	(0.0068)
SBalW	-0.0085*	-0.0100*	0.0104	0.0081
	(0.0047)	(0.0051)	(0.0083)	(0.0068)
BothW	-0.0095**	-0.0108**	0.0150*	0.0052
	(0.0048)	(0.0052)	(0.0084)	(0.0069)
p-values for pairwise comparison				
BothW vs MinW	0.8176	0.0351	0.8975	0.1330
BothW vs SBalW	0.8332	0.8694	0.5865	0.6813
SBalW vs MinW	0.9830	0.0506	0.6776	0.0524
Baseline(Control)	0.086	0.099	0.607	0.207
R^2	0.010	0.011	0.117	0.131
N observations	429,016	429,016	429,016	429,016
N cardholders	25,793	25,793	25,793	25,793

Notes: OLS estimates of Equation 1 using different main payment values or ranges as dependent variables. Columns in Panel A include a sub-sample of debtors who usually failed to pay or paid less than the minimum before the experiment. Columns in Panel B include a sub-sample of debtors who usually paid between the minimum and the statement balance before the experiment. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A8
WARNINGS EFFECT BASED ON PREVIOUS PAYMENT BEHAVIOR, THE MINIMUM AND THE STATEMENT BALANCE AMOUNTS USING INTERACTIONS

Dep Var	Percentage of full (1)	Percentage of full (2)	Percentage of full (3)	Percentage of full (4)	Percentage of full (5)	Percentage of full (6)	Percentage of full (7)	Percentage of full (8)
MinW	0.0013	0.0013	0.0046**	0.0021	0.0049**	0.0022	0.0053**	0.0061**
	(0.0020)	(0.0020)	(0.0023)	(0.0027)	(0.0024)	(0.0029)	(0.0025)	(0.0029)
SBalW	0.0026	0.0014	0.0075***	0.0054**	0.0077***	0.0055*	0.0080***	0.0099***
	(0.0020)	(0.0020)	(0.0023)	(0.0028)	(0.0024)	(0.0030)	(0.0025)	(0.0030)
BothW	0.0041**	0.0037*	0.0078***	0.0060**	0.0081***	0.0065**	0.0090***	0.0130***
	(0.0020)	(0.0019)	(0.0023)	(0.0026)	(0.0024)	(0.0029)	(0.0026)	(0.0030)
X=	*MAD previous % of full (%)*	*MAD previous % of full (%)*	*Diff between SB and min ($)*	*Diff between SB and min ($)*	*Statement Balance ($)*	*Statement Balance ($)*	*Minimum ($)*	*Minimum ($)*
MinW $\times$ X	0.0129	0.0105	-0.0010	0.0017	-0.0009	0.0012	-0.0053	-0.0094
	(0.0159)	(0.0459)	(0.0007)	(0.0018)	(0.0007)	(0.0016)	(0.0036)	(0.0066)
SBalW $\times$ X	0.0312**	0.0658	-0.0004	0.0019	-0.0005	0.0015	-0.0029	-0.0109
	(0.0159)	(0.0462)	(0.0007)	(0.0019)	(0.0007)	(0.0017)	(0.0036)	(0.0070)
BothW $\times$ X	0.0205	0.0276	-0.0005	0.0019	-0.0005	0.0012	-0.0047	-0.0210***
	(0.0158)	(0.0460)	(0.0007)	(0.0017)	(0.0007)	(0.0015)	(0.0037)	(0.0069)
MinW $\times$ X^2		0.0059		-0.0002		-0.0001		0.0022
		(0.1311)		(0.0001)		(0.0001)		(0.0022)
SBalW $\times$ X^2		-0.0905		-0.0002		-0.0001		0.0033
		(0.1317)		(0.0001)		(0.0001)		(0.0024)
BothW $\times$ X^2		-0.0121		-0.0002**		-0.0001*		0.0062***
		(0.1314)		(0.0001)		(0.0001)		(0.0024)
R^2	0.040	0.040	0.040	0.040	0.040	0.040	0,04	0,04
N observation	2,851,884	2,851,884	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939	2,853,939
N cardholders	178,870	178,870	179,706	179,706	179,706	179,706	179,706	179,706

Notes: OLS estimates of Equation 1 plus the main effect of a covariate (X) and its linear and quadratic interaction with the warnings conditions. All models using the percentage paid of the statement balance ("Percentage of Full") as the dependent variable. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A9
COVARIATES FOR EACH QUINTILE - MINIMUM PAYMENT WARNING

	Minimum Payment Warning				
	Q1	Q2	Q3	Q4	Q5
CATE Fraction Paid (%)	-0.3	-0.2	0.0	0.8	1.0
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)
Age (years)	42.0	44.4	44.9	45.5	45.0
	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
Male (%)	48.7	48.9	48.3	47.4	49.1
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)
Mean previous % of full (%)	65.8	80.0	79.8	83.2	70.2
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)
MAD previous % of full (%)	17.0	9.5	11.6	8.9	22.0
	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)
Statement balance ($)	2,420	2,028	1,816	1,776	1,701
	(30)	(20)	(21)	(18)	(22)
Difference between SB and min ($)	1,953	1,547	1,432	1,365	1,371
	(27)	(18)	(18)	(16)	(20)
Previous payment ($)	1,409	1,328	1,144	1,224	919
	(24)	(15)	(11)	(12)	(11)
Monthly income ($)	3,809	3,389	3,689	3,417	3,942
	(25)	(21)	(22)	(20)	(23)
Share of Debt Score (0 to 2.1)	0.9	1.1	1.0	1.1	0.9
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Company's credit score (0-100)	56.0	64.5	67.8	67.8	65.2
	(0.12)	(0.11)	(0.12)	(0.12)	(0.15)
Mean email opening rate (%)	50.0	51.4	51.1	51.6	50.0
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)

Notes: For the minimum-payment warning (vs. control) analysis, quintiles are sorted by the conditional average treatment effects on the fraction paid. Each row shows the mean value for each covariate and standard errors are between parenthesis. For reasons of confidentiality, monetary amounts ($) were re-scaled using a statement balance of $1,949.28 (in US dollars). Share of debt score was re-scaled (0: people who owe other lenders, and 2.14: they owe all their debt to the firm). Credit score was re-scaled where 100 was the best score (lowest risk).

TABLE A10

COVARIATES FOR EACH QUINTILE - STATEMENT BALANCE WARNING

	Statement Balance Warning				
	Q1	Q2	Q3	Q4	Q5
CATE Fraction Paid (%)	-0.3	0.1	0.3	1.0	1.9
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)
Age (years)	44.2	44.6	45.5	43.5	44
	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
Male (%)	50	47.7	50.3	47.8	46.2
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)
Mean previous % of full (%)	78.6	82.7	84.9	70.3	62.4
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)
MAD previous % of full (%)	9.9	8.5	6.7	16.6	27.2
	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)
Statement balance ($)	2,026	1,903	2,064	2,205	1,560
	(29)	(20)	(22)	(23)	(18)
Difference between SB and min ($)	1,646	1,457	1,575	1,755	1,244
	(26)	(18)	(20)	(20)	(16)
Previous payment ($)	1,242	1,280	1,518	1,190	794
	(21)	(13)	(18)	(12)	(10)
Monthly income ($)	3,775	3,487	3,642	3,742	3,553
	(23)	(22)	(22)	(23)	(22)
Share of Debt Score (0 to 2.1)	0.9	1.1	1.1	1.0	0.9
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Company's credit score (0-100)	61.4	66.6	67.3	63.2	62.6
	(0.14)	(0.12)	(0.12)	(0.12)	(0.14)
Mean email opening rate (%)	50.7	51.4	51.9	50.3	49.1
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)

Notes: For the statement balance warning (vs. control) analysis, quintiles are sorted by the conditional average treatment effects on the fraction paid. Each row shows the mean value for each covariate and standard errors are between parenthesis. For reasons of confidentiality, monetary amounts ($) were re-scaled using a statement balance of $1,949.28 (in US dollars). Share of debt score was re-scaled (0: people who owe other lenders, and 2.14: they owe all their debt to the firm). Credit score was re-scaled where 100 was the best score (lowest risk).

TABLE A11

COVARIATES FOR EACH QUINTILE - BOTH WARNINGS

	Both Warnings				
	Q1	Q2	Q3	Q4	Q5
CATE Fraction Paid (%)	-0.2	0.2	0.2	0.9	2.2
	(0.6)	(0.6)	(0.6)	(0.6)	(0.6)
Age (years)	44.5	44.6	42.8	44.5	45.5
	(0.10)	(0.10)	(0.10)	(0.10)	(0.10)
Male (%)	48.4	47.9	50.7	47.8	46.8
	(0.4)	(0.4)	(0.4)	(0.4)	(0.4)
Mean previous % of full (%)	82.4	81.4	73.1	75.3	66.5
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)
MAD previous % of full (%)	8.8	10.4	10.6	14.8	24.4
	(0.1)	(0.1)	(0.1)	(0.1)	(0.1)
Statement balance ($)	2,065	1,743	2,722	1,732	1,514
	(22)	(21)	(26)	(21)	(24)
Difference between SB and min ($)	1,615	1,364	2,133	1,369	1,217
	(20)	(19)	(24)	(19)	(21)
Previous payment ($)	1,514	1,153	1,774	944	686
	(18)	(14)	(22)	(11)	(9)
Monthly income ($)	3,582	3,456	4,053	3,511	3,641
	(22)	(21)	(25)	(21)	(23)
Share of Debt Score (0 to 2.1)	1.1	1.1	1.0	1.0	0.8
	(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
Company's credit score (0-100)	66.6	66.3	60.8	64	63.6
	(0.12)	(0.12)	(0.13)	(0.12)	(0.14)
Mean email opening rate (%)	51.6	50.9	51.7	50.0	49.6
	(0.2)	(0.2)	(0.2)	(0.2)	(0.2)

Notes: For the both warning (vs. control) analysis, quintiles are sorted by the conditional average treatment effects on the fraction paid. Each row shows the mean value for each covariate and standard errors are between parenthesis. For reasons of confidentiality, monetary amounts ($) were re-scaled using a statement balance of $1,949.28 (in US dollars). Share of debt score was re-scaled (0: people who owe other lenders, and 2.14: they owe all their debt to the firm). Credit score was re-scaled where 100 was the best score (lowest risk).

A12. Material of online experiment

- Are you more familiar with U.S. dollars or British pounds?
  - $ (US Dollar)
  - £ (British Pound)
  - I am not familiar with any of these currencies.

- Payment information

Imagine that you own a credit card and you are about to make your monthly repayment. The next screen will show summary information from your credit card balance.

Please consider how much you can afford to pay, and treat your payment decision as you would in your everyday life.

Payment Due Date	[date]
Statement Balance	[$]
Minimum Payment	[$]
[Warnings, if any]	

[Values depending on the currency selected:]
UK: minimum (£17.30), statement balance (£691.82).
US: minimum ($33.89), statement balance ($1,355.67).
[Random assignment to no warnings, minimum payment, statement balance, or both warnings, using:]

Minimum payment warning: "If you make at least the Minimum Payment ([$]), **you will pay additional interest, but you will avoid late fees.**"

Statement balance warning: "If you pay the Statement Balance [$]), **you will not pay any additional interest or fees.**"

- How much will you repay?
  - Minimum payment
  - Statement balance
  - Other amount: [participants could type in the amount]

- Please answer these two questions about the information you just saw [the following two questions are randomly ordered]
  - How well do you understand what the minimum payment means? (1=Not at all/7=Very well)

◦ How well do you understand what the statement balance means? (1=Not at all/7=Very well)

• During the last 12 months, please answer to what extent you agree with the following statements [1 “Strongly disagree”, 2 “Disagree”, 3 “Somewhat disagree”, 4 “Neither disagree nor agree”, 5 “Somewhat agree”, 6 “Agree”, 7 “Strongly agree”]
◦ I always pay my credit card balance off in full each month
◦ I often make only the minimum payment on my credit card bills

• [Payments with credit card from Salisbury (2014)]

• Considering the last 12 months, please answer which range best reflects the amount you usually repay for your credit card [using UK and US ranges]

• [Financial vulnerability from Salisbury and Zhao (2020)]

• [Financial literacy from Navarro-Martinez et al. (2011)]

• [Cognitive reflection test from Frederick (2005)]

• Demographics: age, gender, occupation, annual income [using UK and US ranges], and educational level. Were the rules about the study clear? Do you have any comments about the study?

• Open-ended question about any comment about the study.

TABLE A13 - A
WARNINGS EFFECT BASED ON COGNITIVE FACTORS
IN THE ONLINE EXPERIMENT (WITH INTERACTIONS)

Dep. Var.	Percentage of Full (1)	Percent of Full (2)	Percentage of Full (3)
MinW	0.047	-0.067	-0.010
	(0.065)	(0.125)	(0.100)
SBalW	0.171***	0.110	0.198*
	(0.063)	(0.122)	(0.097)
BothW	0.246***	0.209*	0.346***
	(0.062)	(0.125)	(0.097)
$X =$	*Financial Vulnerability*	*Financial Literacy*	*CRT (Score)*
X	-0.027	0.097**	0.074**
	(0.018)	(0.039)	(0.033)
MinW $\times$ X	-0.035	0.043	0.018
	(0.027)	(0.057)	(0.048)
SBalW $\times$ X	-0.025	0.027	-0.019
	(0.028)	(0.055)	(0.049)
BothW $\times$ X	-0.035	0.013	-0.067
	(0.026)	(0.057)	(0.048)
R^2	0.115	0.130	0.085
N observations	400	400	400

Notes: OLS estimates of Equation 1 plus the main effect of a covariate (X) and its linear interaction with the warnings conditions. All models using the percentage of the statement balance paid ("Percentage of Full") as the dependent variable. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

TABLE A13 - B

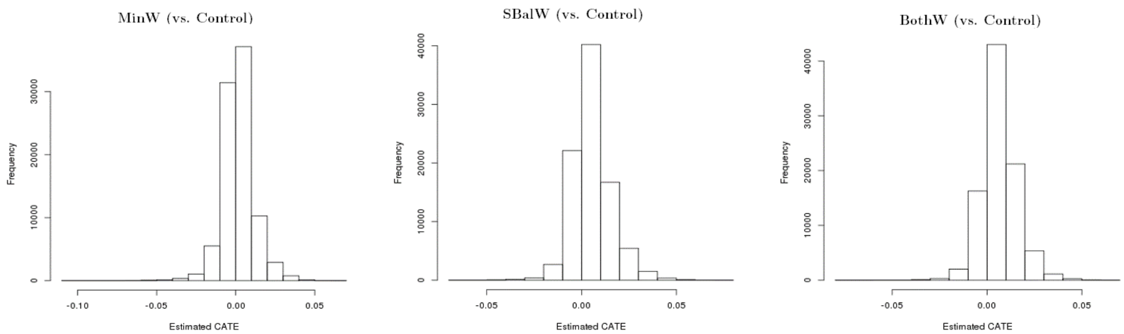
WARNINGS EFFECT BASED ON COGNITIVE FACTORS IN THE ONLINE EXPERIMENT

Subsample	*High Financial Vulnerability*	*Low Financial Vulnerability*	*Low Financial Literacy*	*High Financial Literacy*	*Low CRT (Score)*	*High CRT (Score)*
Dep. Var.	Percentage of Full (1)	Percentage of Full (2)	Percentage of Full (3)	Percentage of Full (4)	Percentage of Full (5)	Percentage of Full (6)
MinW	0.037	0.003	-0.051	0.134*	0.030	0.003
	(0.067)	(0.111)	(0.081)	(0.075)	(0.094)	(0.076)
SBalW	0.130**	0.207*	0.139*	0.185**	0.205**	0.135*
	(0.065)	(0.117)	(0.082)	(0.072)	(0.093)	(0.076)
BothW	0.183***	0.268**	0.195**	0.319***	0.331***	0.154*
	(0.063)	(0.125)	(0.079)	(0.075)	(0.090)	(0.078)
p-values from pairwise comparison						
BothW vs MinW	0.027	0.030	0.002	0.021	0.001	0.064
BothW vs SBalW	0.401	0.628	0.479	0.081	0.143	0.814
SBalW vs MinW	0.166	0.072	0.018	0.500	0.055	0.095
R^2	0.036	0.068	0.051	0.110	0.094	0.030
N observations	286	114	245	155	183	217

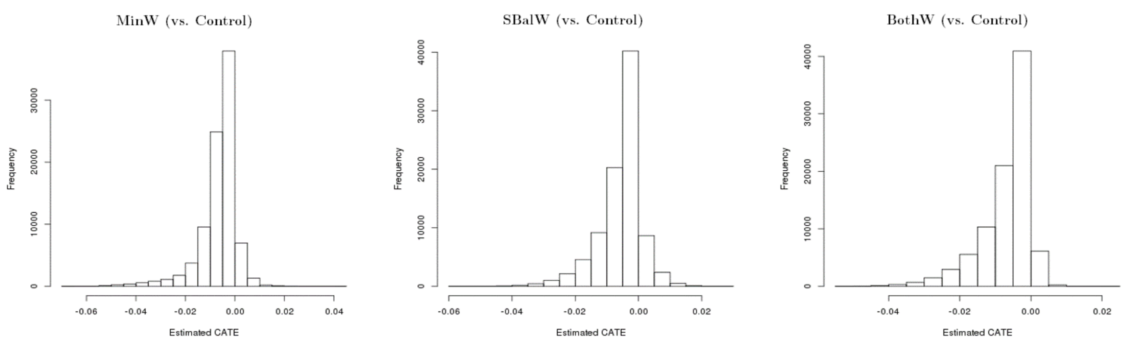
Notes: OLS estimates of Equation 1 using the percentage of the statement balance paid ("Percentage of Full") as the dependent variable. Each column is a different sub-sample. SEs between parenthesis. MinW is the Minimum warning, SBalW is the Statement Balance Warning, and BothW is the Both Warnings condition.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

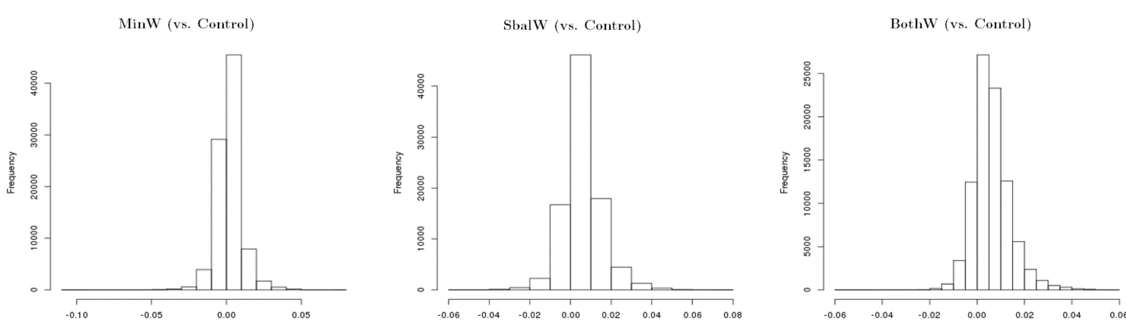
FIGURE A1. DISTRIBUTIONS OF HETEROGENEOUS TREATMENT EFFECTS FOR EACH WARNING USING THE CONTROL AS THE BASELINE



(a) Full



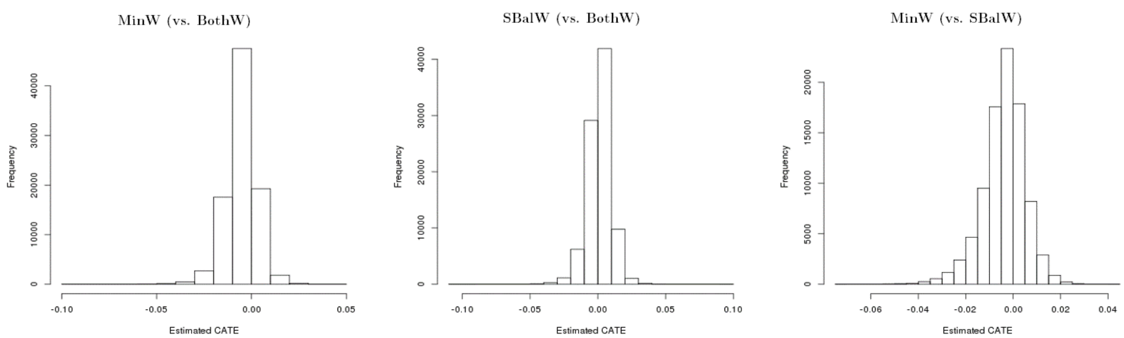
(b) Less than Min



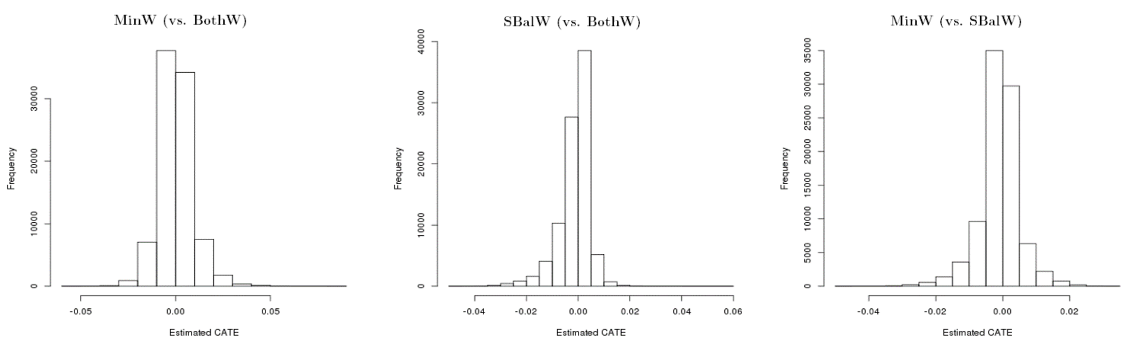
(c) Percent of Full

Notes: Histograms of the causal random forests. Panels (a), (b) and (c) indicate the dependent variable. The baseline is the control group for all the graphs.

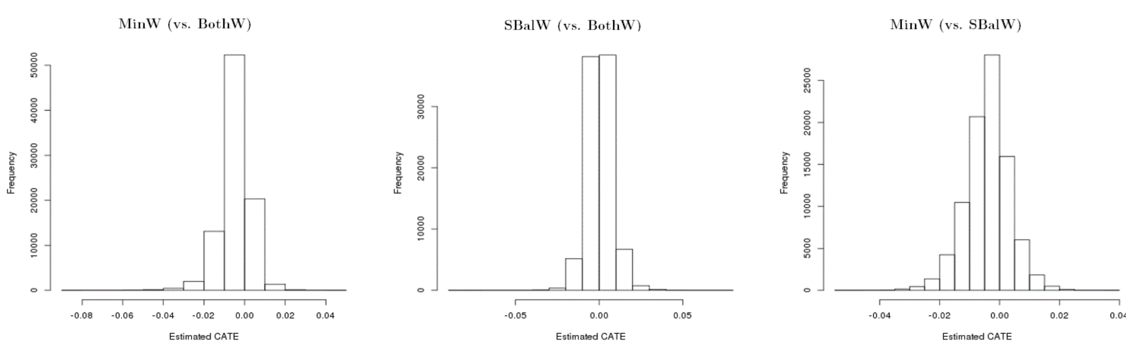
FIGURE A2. DISTRIBUTIONS OF HETEROGENEOUS TREATMENT EFFECTS BETWEEN WARNING CONDITIONS



(a) Full



(b) Less than Min



(c) Percent of Full

Notes: Histograms of the causal random forests. Panels (a), (b) and (c) indicate the dependent variable. The baseline is the second group after "vs." in each graph.